# CardamomOT — analysis report

/Users/eventre/Documents/GitHub/CardamomOT/experimental_datasets/Schiebinger

Generated on 2026-10-15 10:54

## Run

| | |
|---|---|
| **Split** | train |
| **Stimulus penalisation** | 1.0 |
| **Prior penalisation** | 1.0 |
| **Genes** | 108 |
| **Stimuli** | 2 |
| **Observed cells** | 5618 |
| **Timepoints** | 0, 12, 24, 36, 48, 60, 72, 84, 96, 108, 120, 132, 144, 156, 168, 180, 192, 204, 216, 228, 240, 252, 264, 276, 288, 300, 312, 324, 336, 348, 360, 372, 384, 396, 408, 420, 432 |
| **Cell types** | MEF, Neural, Stromal, Trophoblast, iPSC, Epithelial |
| **Network shown** | inter_simul.npy |
| **Embeddings** | UMAP; mRNA: log1p=True, normalise=False, cell_depth_for_representation=True (observed / depth factor, model draws at the reference depth) |

## Perturbations (perturbation_simulation)

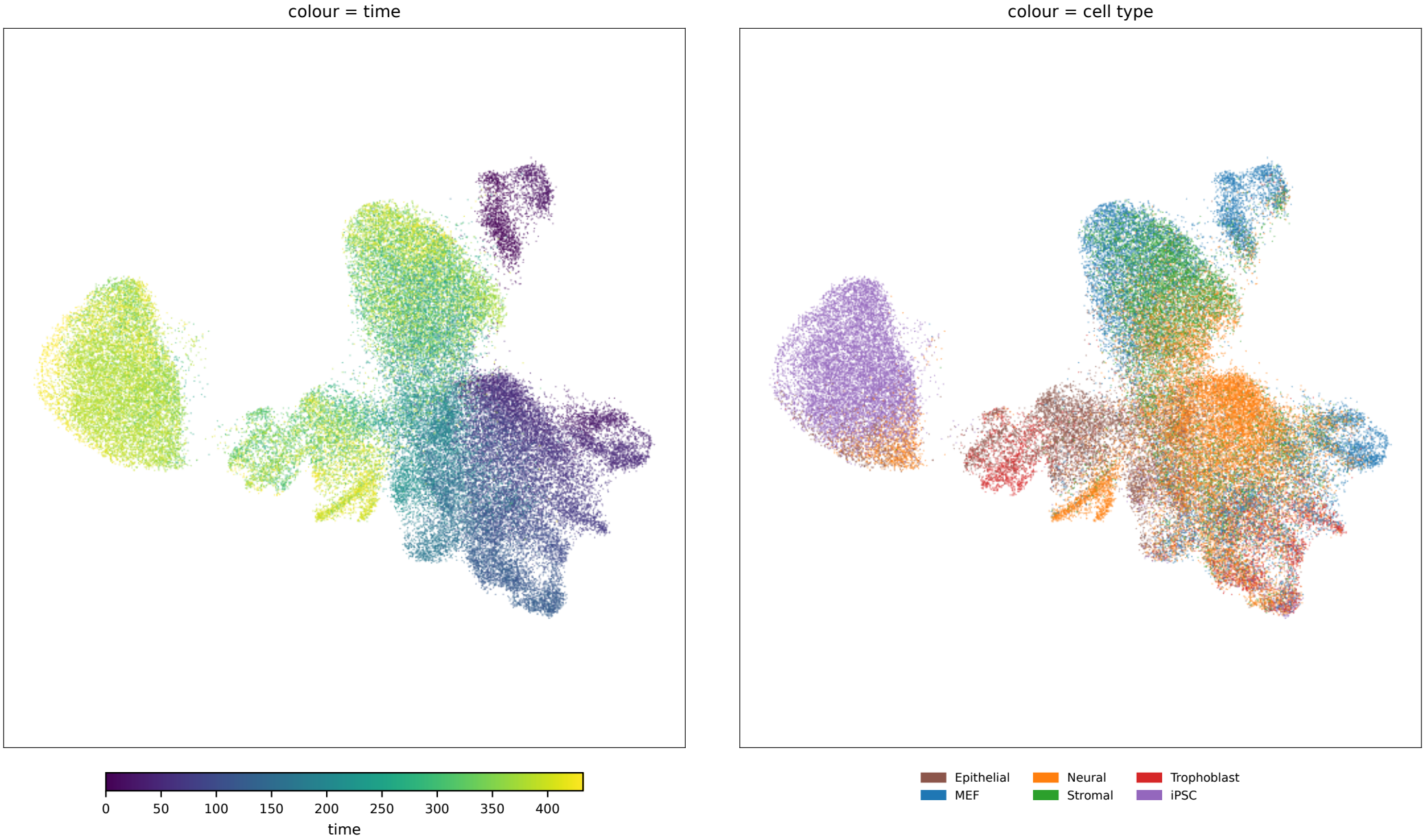
- ✓ P1 OV Obox6
- ✓ P2 OV Zfp42

## Contents


Data — UMAP of all the cells of Data/data.h5ad (time, samples); teaser: classical OT vs CardamomOT displacement fields (all times, then per interval), then cell-type transitio
1. Generative model — UMAPs of data, trajectories, NB mixture, network modes and simulation; cell-type proportions; gene-pair correlations; proteins
2. Gene regulatory network — regulatory power (violin plots), top-20 regulators + stimulus and top-20 regulated genes (mutual top-10 edges)
3. In-silico perturbations — overview across KO/OV, then one page per perturbation
4. Proliferation — prior vs learned net rates, population growth, proteins driving growth; protein dilution at the birth rate
5. Learned dynamics — mRNA and protein fields (mechanistic velocity, displacement along trajectories), summary on mRNA
6. Held-out test cells — predictions with the network fixed vs the test data

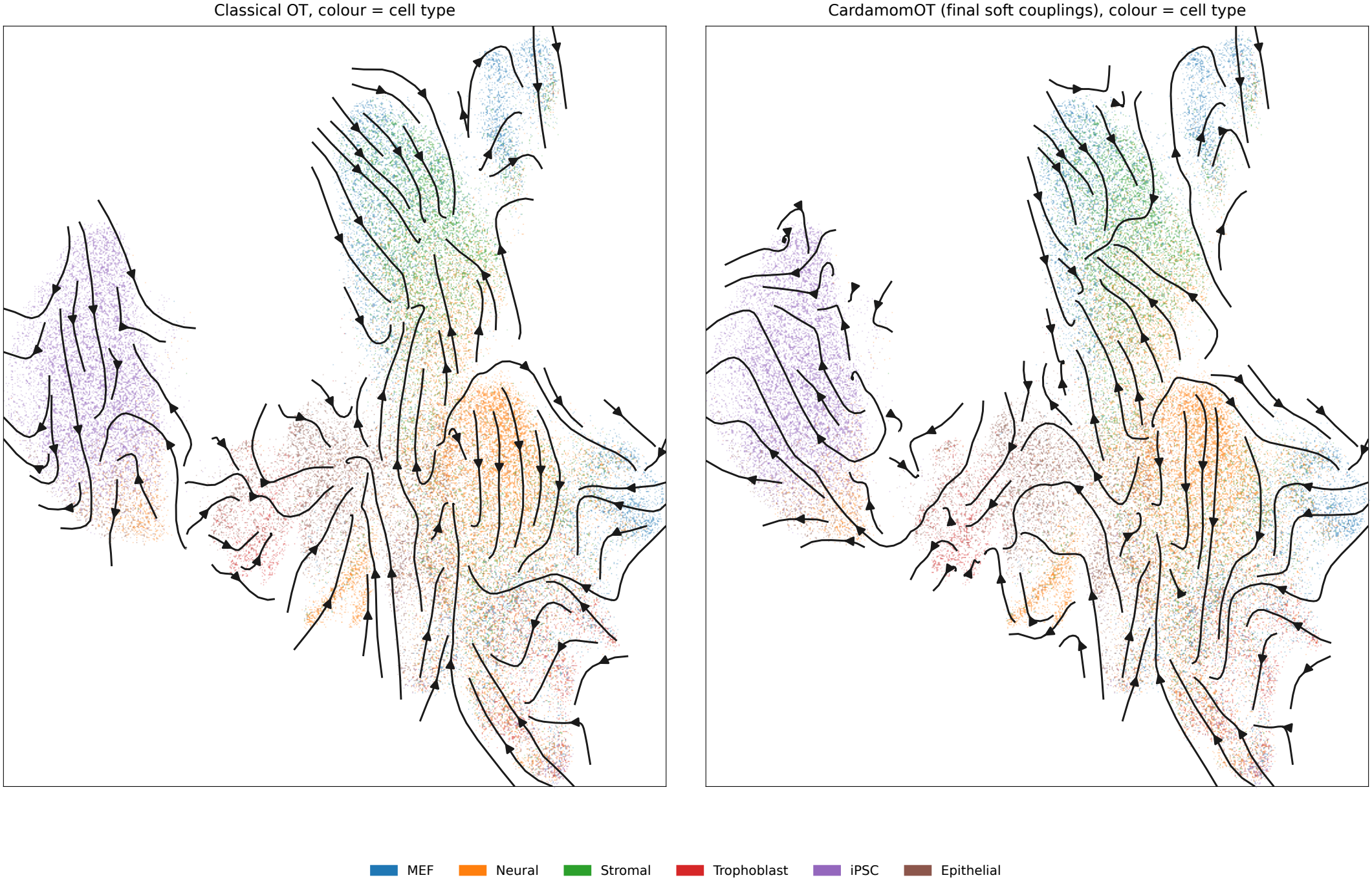
# Data — UMAP of all the cells

Data/data.h5ad, 53678 cells: log1p(counts), 108 highly variable genes, PCA (50 components), kNN graph, UMAP. Stored in cardamomOT/embedding_data.npz.



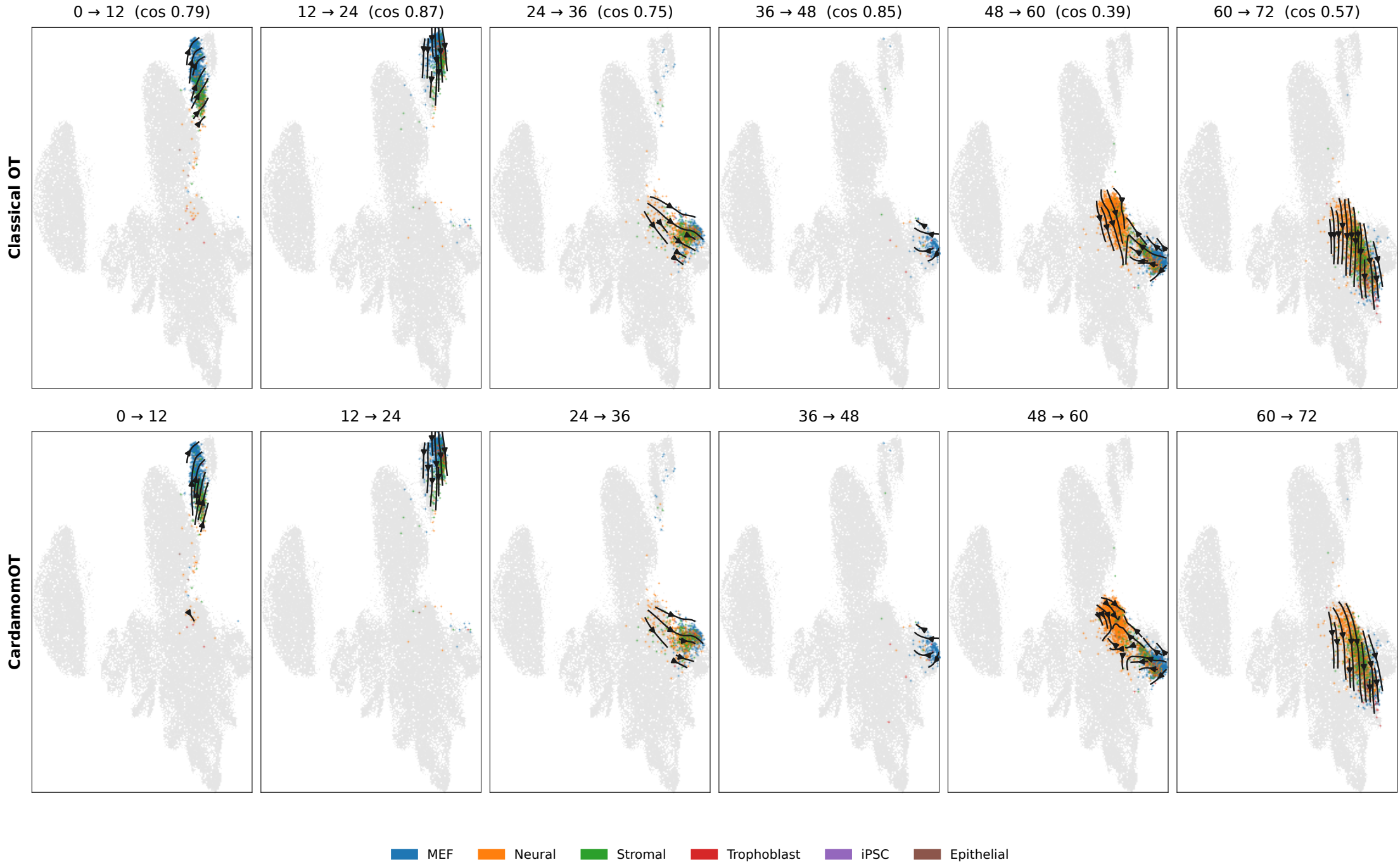
# Teaser — displacement fields of classical OT and CardamomOT

Displacement to the mean descendant (x(t+1) − x(t), not divided by Δt), all times. Agreement on the cells with both direct displacements: weighted cosine 0.60 (PCA space). CardamomOT: final soft couplings of the trajectories (5618 cells); cells never reached or outside the training set: Gaussian kernel on the model mRNAs.



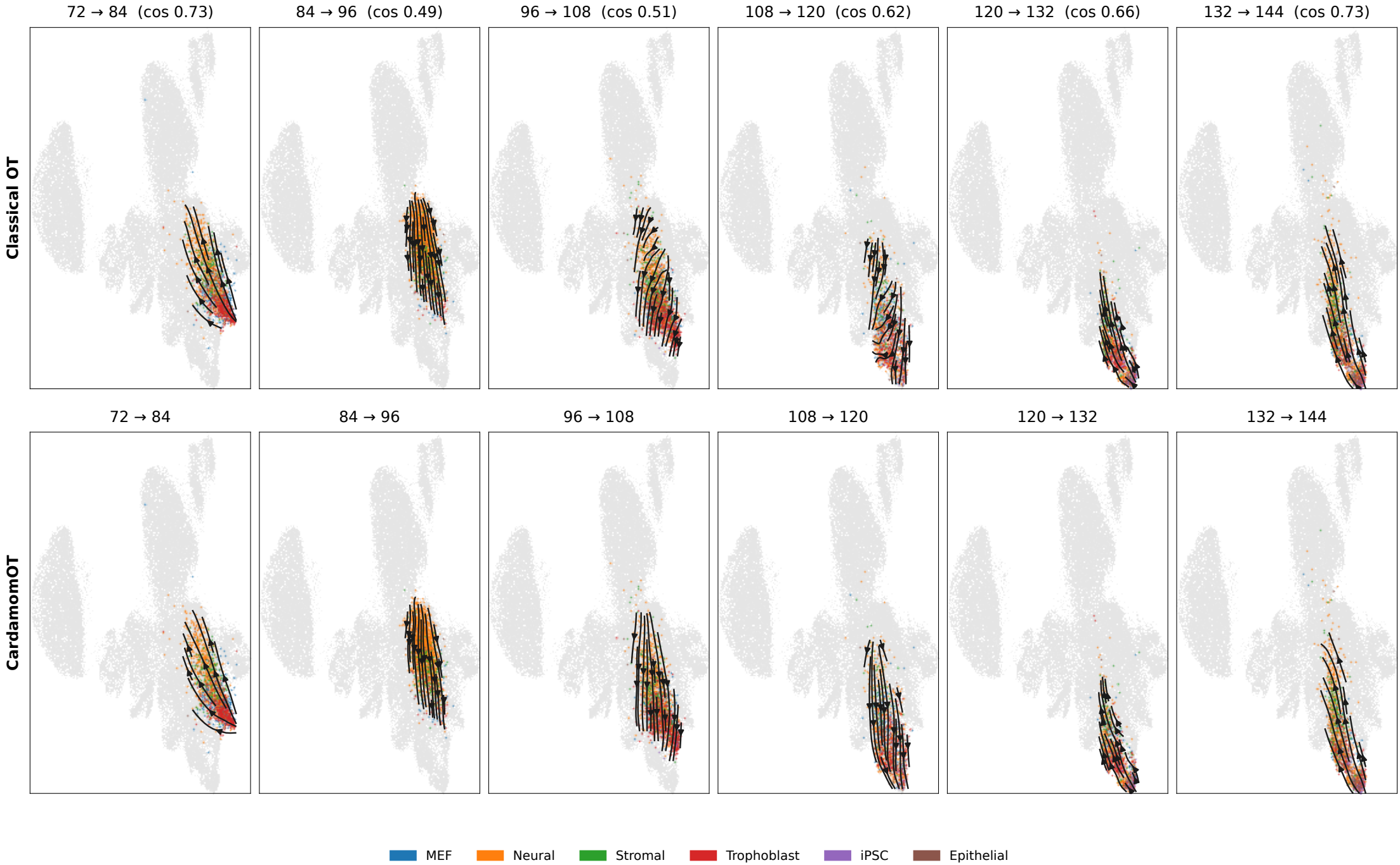
# Teaser — displacement fields over time, classical OT vs CardamomOT

Cells of each t_from (coloured; other cells in grey) towards their mean descendant at t_to, kNN projection among the cells of t_from and t_to. cos: weighted cosine between the two methods on the cells of t_from with both direct displacements (PCA space).



# Teaser — displacement fields over time, classical OT vs CardamomOT

Cells of each t_from (coloured; other cells in grey) towards their mean descendant at t_to, kNN projection among the cells of t_from and t_to. cos: weighted cosine between the two methods on the cells of t_from with both direct displacements (PCA space).



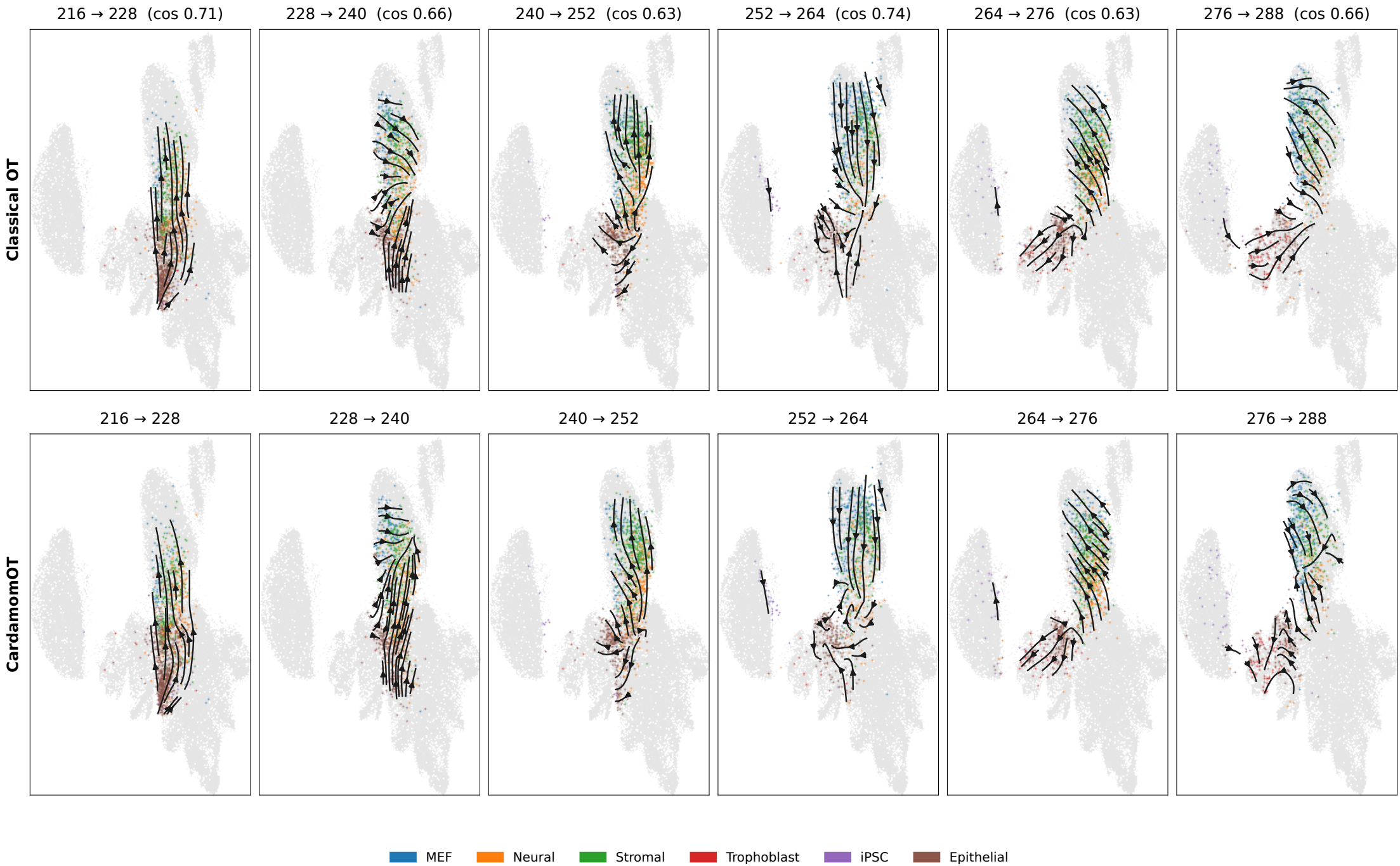
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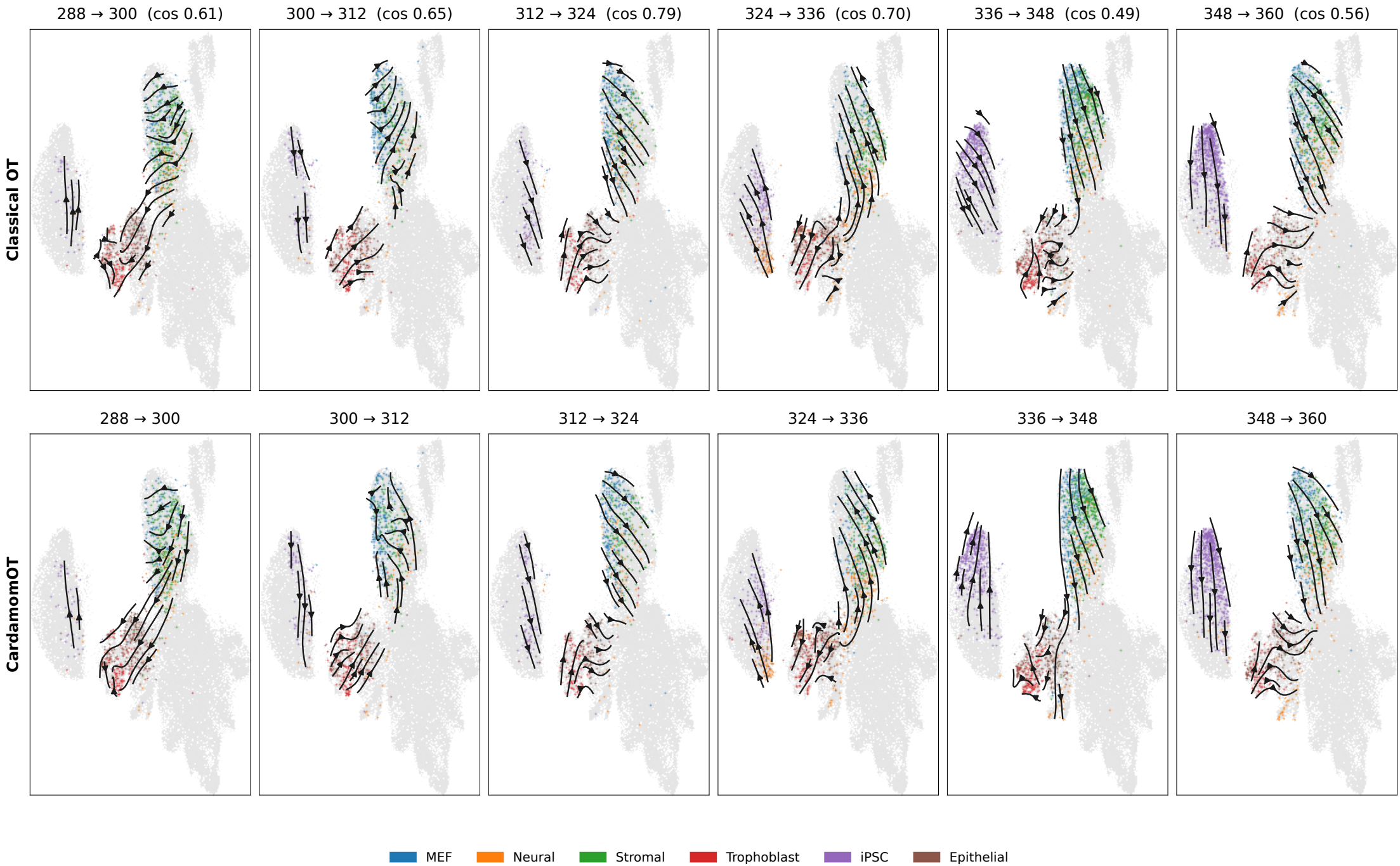
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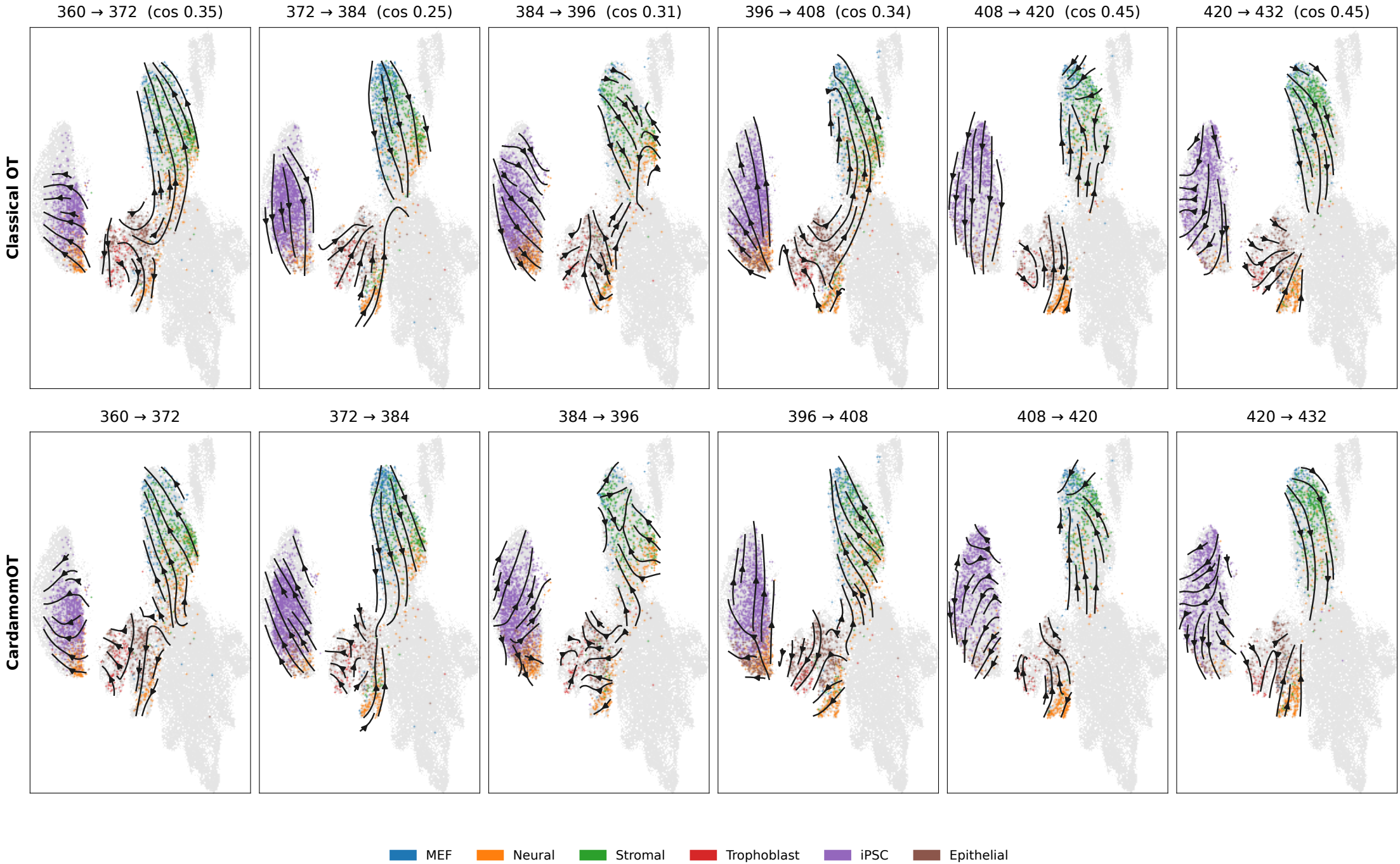
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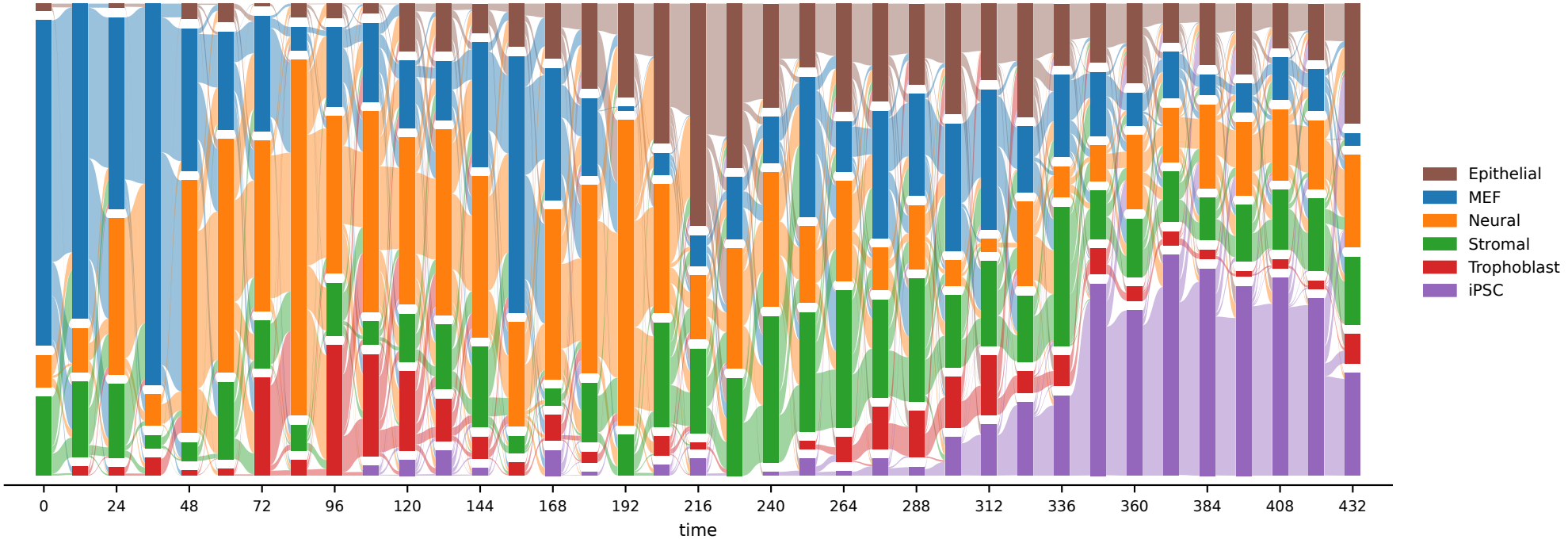
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# Classical OT — cell-type transitions and fate genes

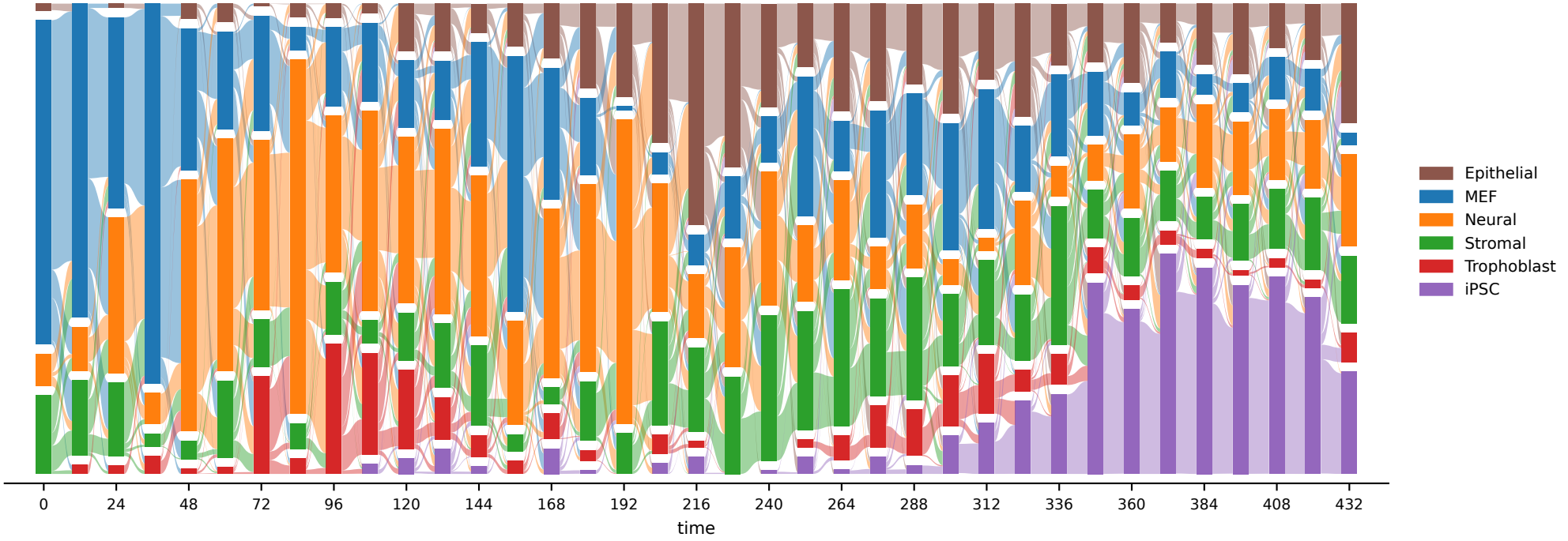
Waddington-OT couplings (every gene, train cells, per sample, growth from the net rate), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (classical_OT/fate_genes.csv).




| | Top-10 genes up in the ancestors (vs ancestors of the other cells) | Top-10 genes up in the cell type (vs the other cells) |
|---|---|---|
| Epithelial | **Krt8** **Epcam** **Cldn7** **Hspb1** **Bbx** **Marcksl1** **Msx2** Chchd10 Gata2 Dmrtc2 | **Krt8** **Epcam** **Cldn7** **Hspb1** **Bbx** **Msx2** Wnt7b **Marcksl1** Fos Psme1 |
| MEF | **Col1a2** **Thbs1** **Lox** **S100a6** **Bgn** **Prss23** **Tnfrsf12a** Mgp **Ncam1** Vcam1 | **S100a6** **Col1a2** **Thbs1** **Prss23** **Lox** **Tnfrsf12a** **Bgn** Prkcdbp **Ncam1** Nmt1 |
| Neural | **S100a6** **2810417H13Rik** **Fam162a** **Xist** **Id3** **Tmem35** **Gng12** Ggh Prkcdbp **Nme2** | **Xist** **2810417H13Rik** **Id3** **S100a6** **Fam162a** **Gng12** Dmrtc2 Ccl7 **Nme2** **Tmem35** |
| Stromal | **Mgp** **Bgn** **Col1a2** **Gabarap** **Thbs1** **Id3** **Cdkn2a** **Lox** **Xist** Cdkn2b | **Mgp** **Col1a2** **Bgn** **Gabarap** **Lox** **Xist** **Thbs1** **Id3** S100a6 **Cdkn2a** |
| Trophoblast | **Sri** **Krt8** **Gng12** **Hmgn5** Gata2 **Bbx** Batf3 Cited2 Elf3 **Pparg** | **Sri** **Gng12** S100a6 **Hmgn5** **Krt8** Cyb5r3 Gstp1 **Bbx** **Pparg** AY036118 |
| iPSC | **Dppa5a** **Chchd10** **Mylpf** **Fabp3** **Mybl2** **Ube2s** **Cdc5l** **Sox2** **Zfp42** Mrps24 | **Dppa5a** **Chchd10** **Fabp3** **Mylpf** **Mybl2** **Ube2s** **Cdc5l** Fmr1nb **Zfp42** **Sox2** |

# Teaser — CardamomOT: cell-type transitions and fate genes

Final soft couplings of the CardamomOT trajectories (model genes, data_<split> cells), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (cardamomOT/teaser_fate_genes.csv).




| | Top-10 genes up in the ancestors (vs ancestors of the other cells) | Top-10 genes up in the cell type (vs the other cells) |
|---|---|---|
| Epithelial | **Krt8** **Epcam** **Cldn7** **Hspb1** **Bbx** **Msx2** **Wnt7b** Gadd45b Marcksl1 Dmrtc2 | **Krt8** **Epcam** **Cldn7** **Hspb1** **Wnt7b** **Msx2** **Bbx** Fos Pdgfb Psme1 |
| MEF | **Col1a2** **S100a6** **Prss23** **Thbs1** **Lox** Bgn **Tnfrsf12a** **Prkcdbp** Ncam1 Mgp | **S100a6** **Col1a2** **Prss23** **Thbs1** **Tnfrsf12a** Xist **Lox** **Prkcdbp** Nmt1 Gng12 |
| Neural | **2810417H13Rik** **Dmrtc2** **Hist1h2ap** **Tmem35** **Nme2** Myc Gm26917 S100a6 Ggh Fam162a | **2810417H13Rik** **Dmrtc2** Id3 Nr2f1 Pou3f1 **Nme2** **Tmem35** Pitx2 **Hist1h2ap** Ccl7 |
| Stromal | **Mgp** **Bgn** **Col1a2** **Gabarap** **Thbs1** **Id3** Cdkn2b **Cdkn2a** Nfic Txnip | **Mgp** **Col1a2** **Bgn** **Gabarap** Xist Lox **Thbs1** **Id3** S100a6 **Cdkn2a** |
| Trophoblast | **Sri** **Gng12** **Krt8** **Hmgn5** Bbx Cited2 Gata2 Batf3 Cldn7 Ube2s | **Gng12** **Sri** S100a6 **Hmgn5** Cyb5r3 Xist Gm26917 AY036118 Fam162a **Krt8** |
| iPSC | **Dppa5a** **Chchd10** **Fabp3** **Mylpf** **Mybl2** **Sox2** **Zfp42** **Cdc5l** **Ube2s** Tcl1 | **Dppa5a** **Chchd10** **Fabp3** **Mylpf** **Mybl2** **Ube2s** **Cdc5l** **Zfp42** **Sox2** Fmr1nb |

# Genes of the model (108)

Data/data_train.h5ad, alphabetical order.

1110008F13Rik
2810417H13Rik
Ascl2
AY036118
Batf3
Bbx
Bgn
Bhlhe40
Blvrb
Ccl7
Cdc5l
Cdkn2a
Cdkn2b
Cenpm
Chchd10
Cited2
Cldn7
Col1a2
Csrp2
Ctsl
Cyb5r3
Ddit3
Dmrtc2
Dppa4
Dppa5a
Dynll2
Ecscr
Egr1
Elf3
Epcam
Erdr1
Ereg
Esrrb
Fabp3
Fam162a
Fam20c
Fmr1nb
Fos
Gabarap
Gadd45b
Gata2
Gata3
Ggh
Gm26917
Gng12
Gstp1
Hes1
Hesx1
Hist1h2ap
Hmgn5
Hspb1
Id3
Klf2
Klf4
Krt8
Lncenc1
Lox
Marcksl1
Mgp
Mrps24
Msx2
Mybl2
Myc
Mycn
Mylpf
Ncam1
Nfic
Nme2
Nmt1
Nr2f1
Nupr1
Obox6
Ovol2
Pdgfb
Pitx2
Pou3f1
Pparg
Prkcdbp
Prr5
Prss23
Psme1
Rala
S100a1
S100a6
Scand1
Sdc4
Shisa8
Snhg6
Snhg9
Sox2
Spic
Sri
Tax1bp3
Tcl1
Thbs1
Tmem35
Tnfrsf12a
Tomm40
Txnip
Ube2s
Vapa
Vbp1
Vcam1
Vps8
Wnt7b
Xist
Ypel3
Zfp42

# 0. Per-cell sequencing depth

Diagnostic on the whole transcriptome within homogeneous groups (sample, time, cell type); counts are modelled as NB(k, c / s_i) when the depth factor s_i is applied.

**Depth**

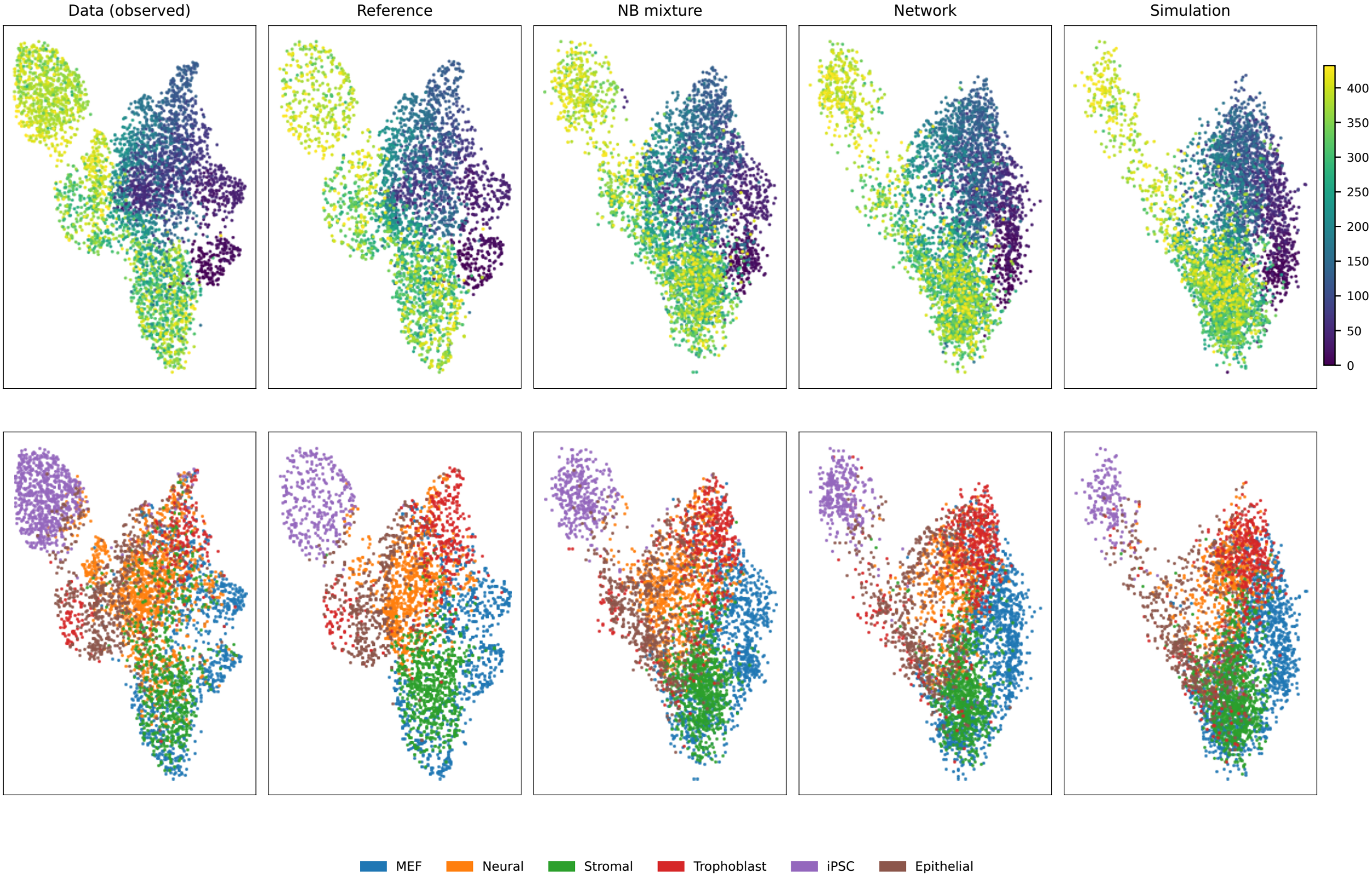
not estimated

**Reason**

108 genes < 10000

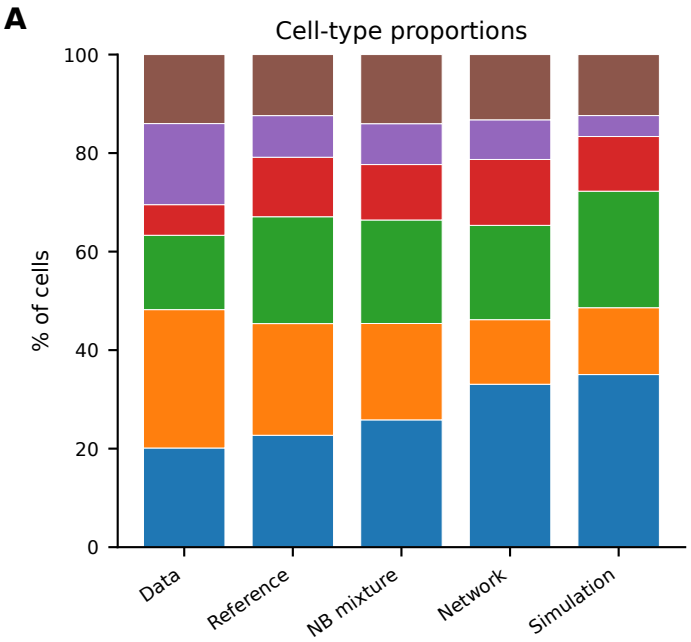
# 1. Generative model — trajectories and simulation

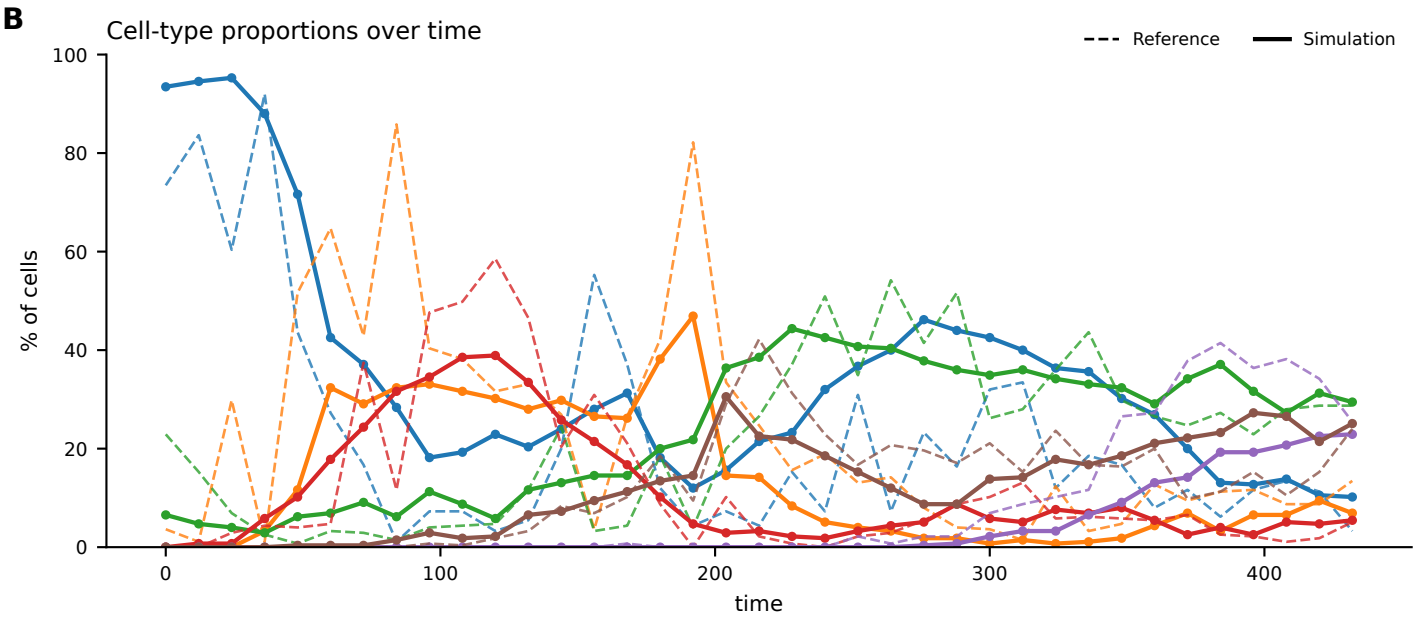
UMAP learned on the observed data; the trajectories (Reference: one state per ancestor, i.e. without proliferation, as the simulation), the NB mixture and network-driven modes along them and the full simulation are projected onto it.

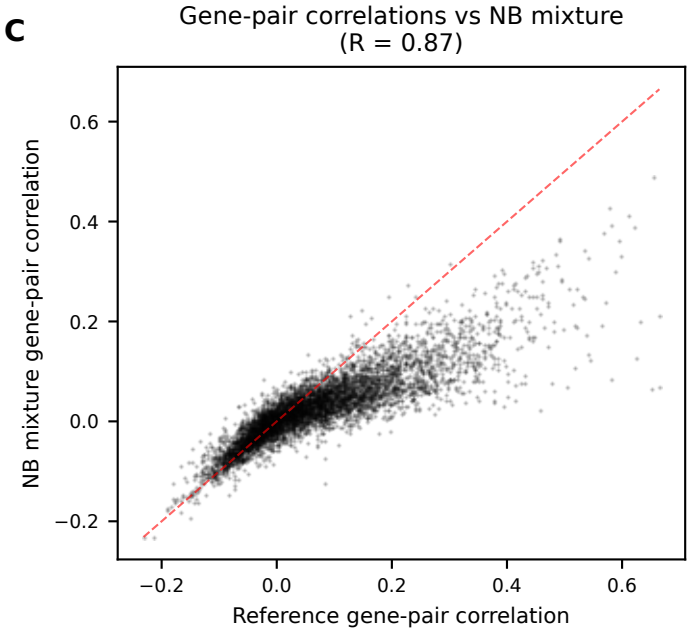


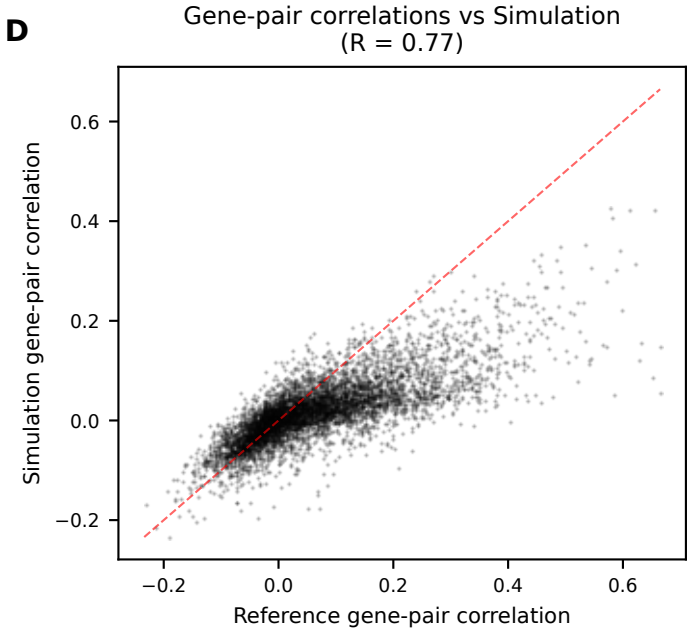
# 1. Generative model — quantitative checks

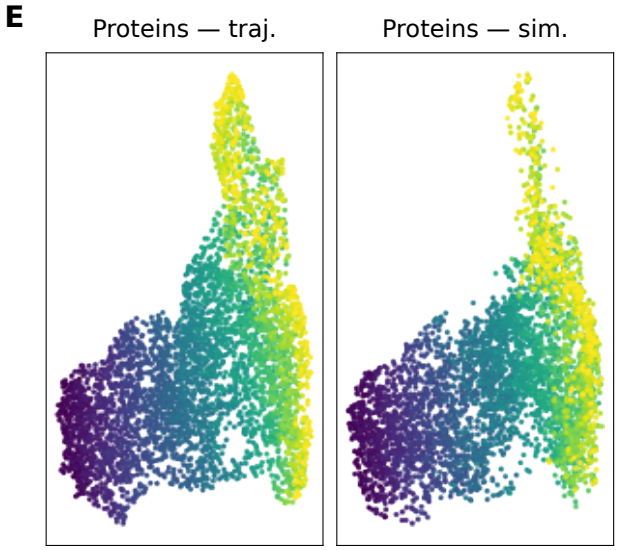
Reference = RNA of the trajectories equivalent to the simulation (without proliferation).



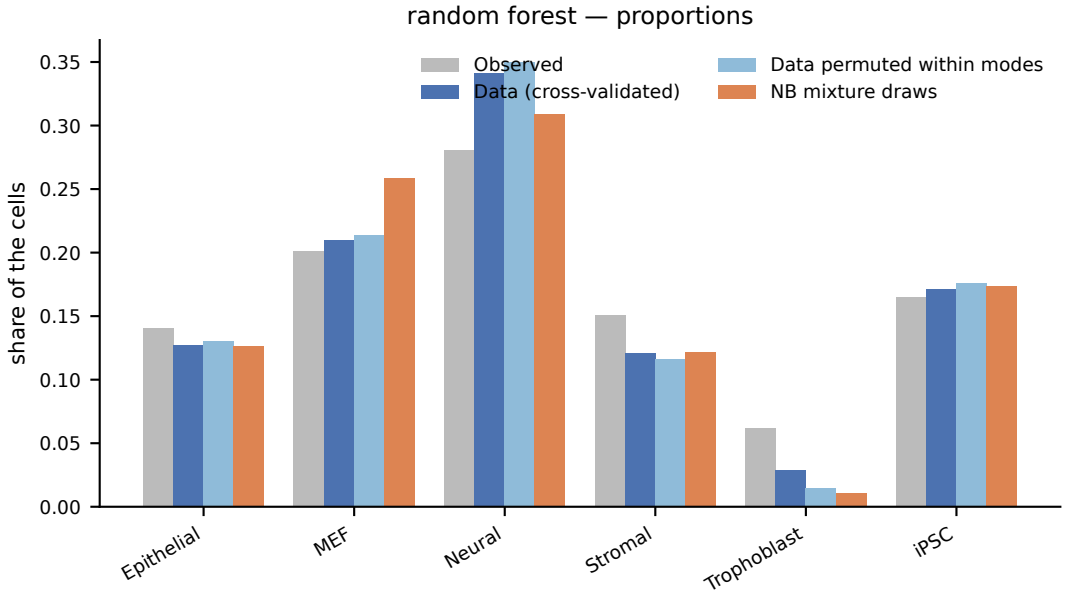



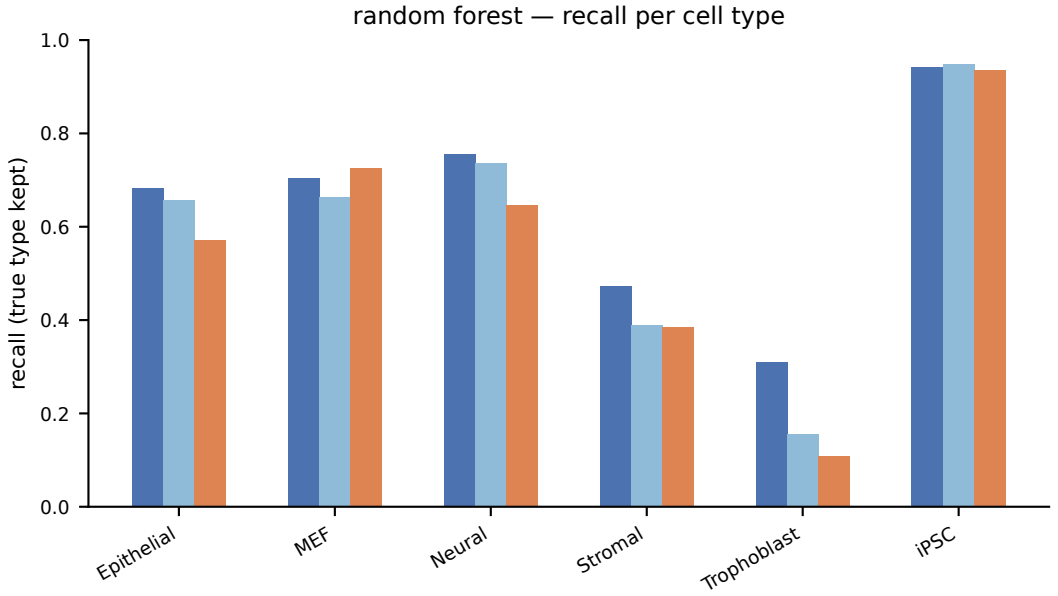



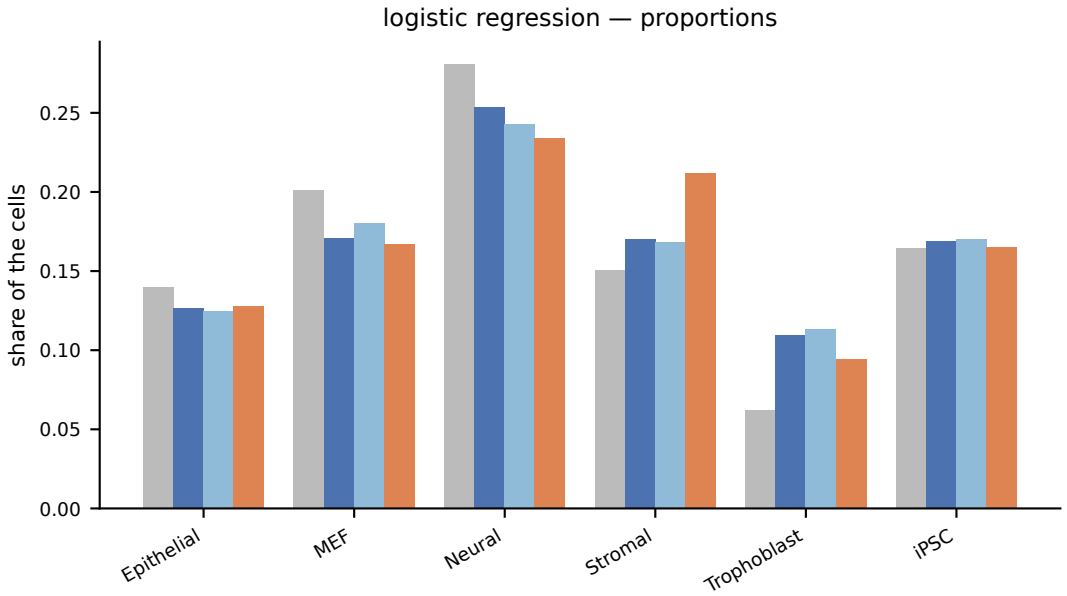


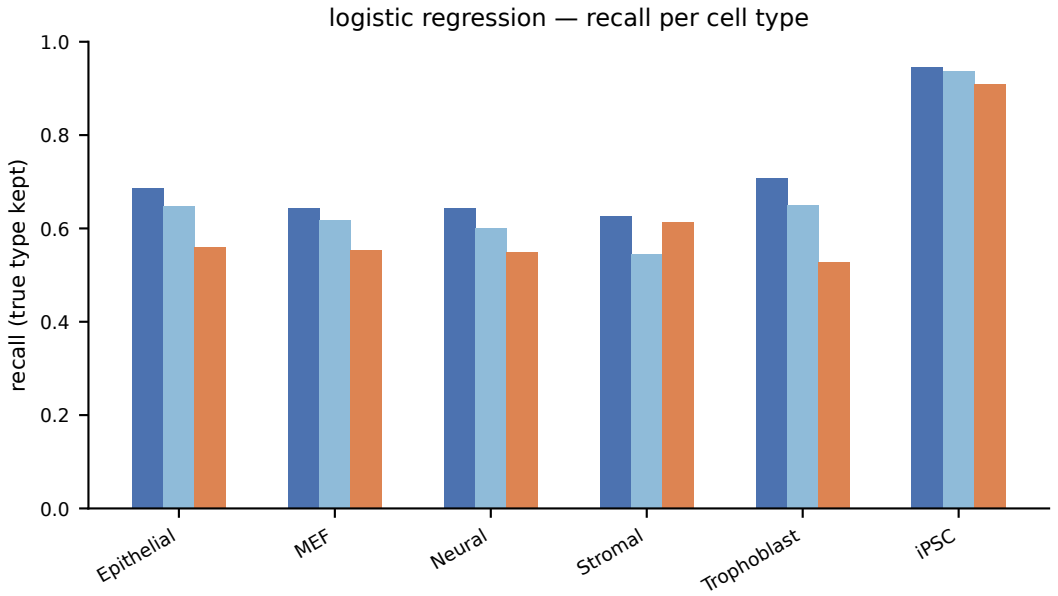

# 1. Generative model — cell-type identity in the NB mixture

One draw per real cell, per-sample classifiers trained on the other cells. Cross-validated data = classifier limit; permuted within modes (same sample, time, mode) = best model with these modes. Gap data → permuted: modes too coarse; permuted → draws: NB fit.



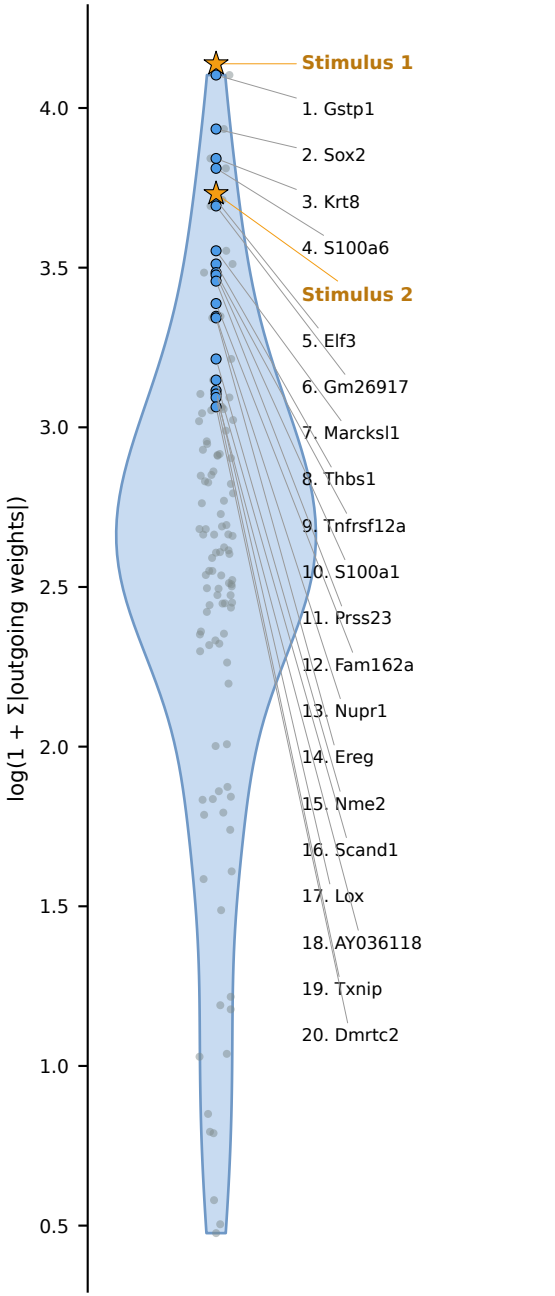



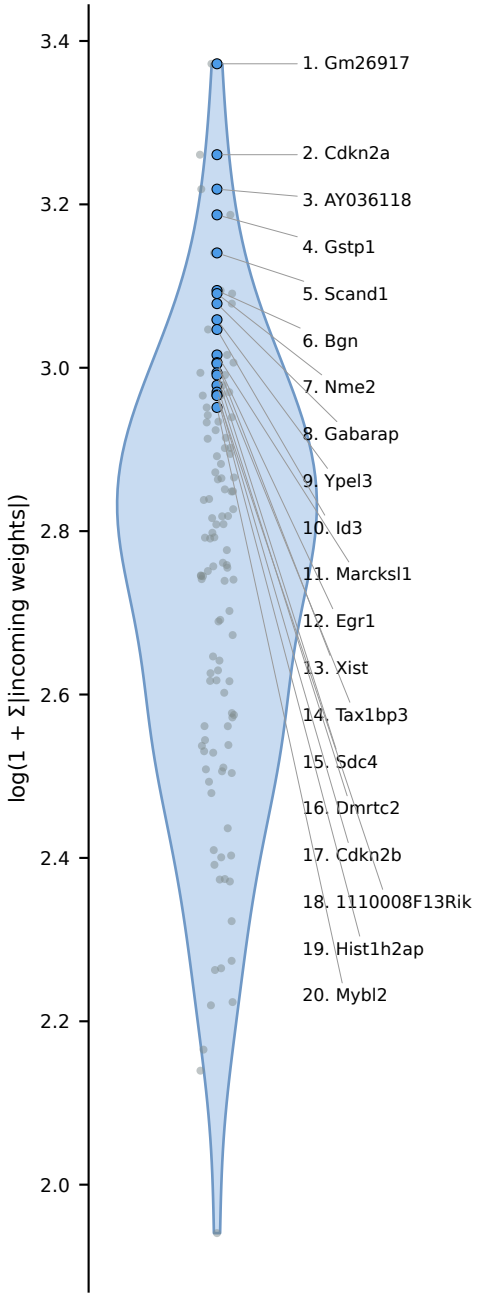
# 2. Gene regulatory network — regulatory power

Self-interactions excluded. Top 20 genes labelled; stimulus shown as a star (stimulus = 1.0 ≥ 0.5).

**A**




**B**




**C** Top 20 regulators

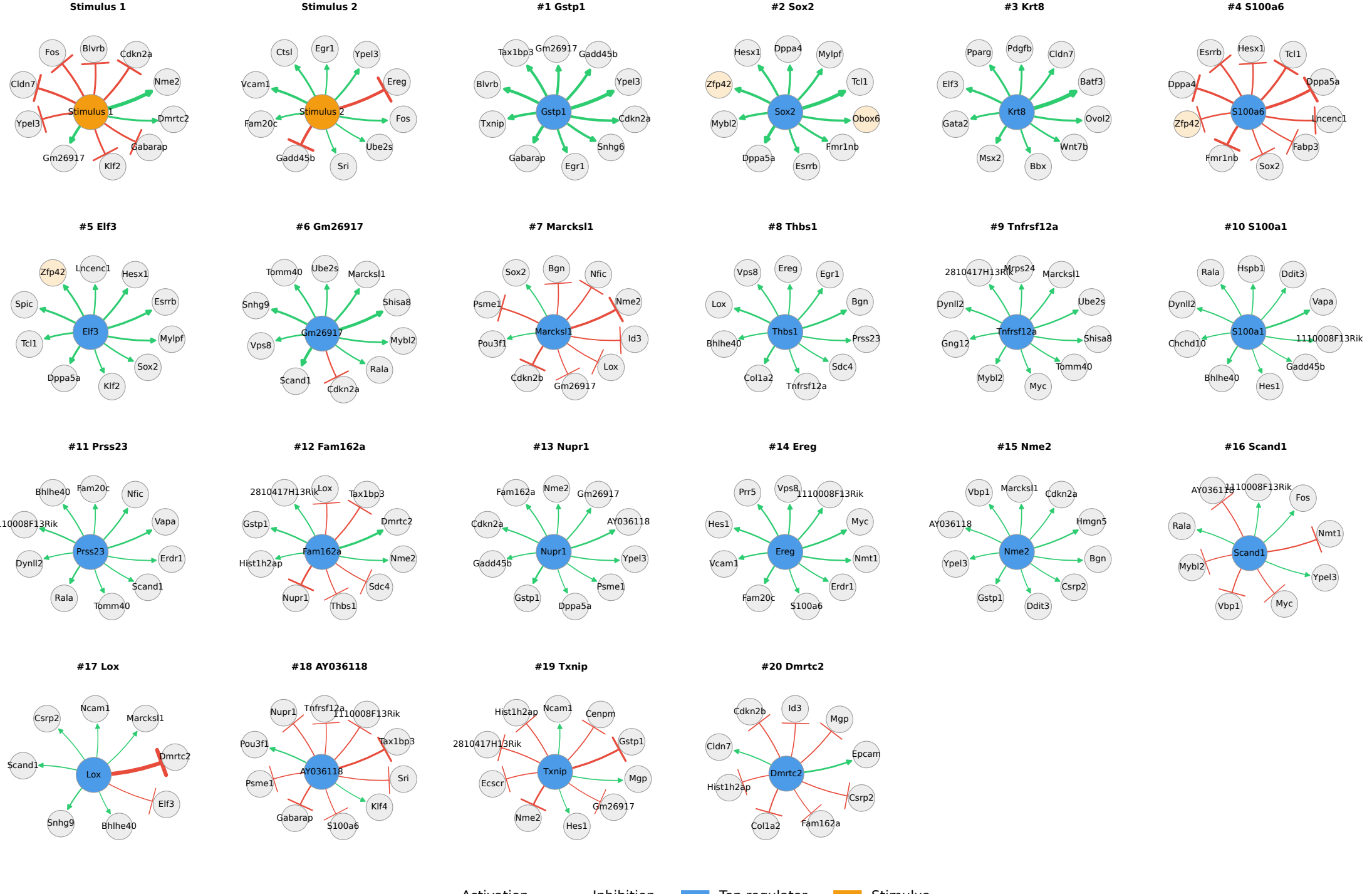
| # | Regulator | Σ\|w\| | act. | inh. |
|---|---|---|---|---|
| 1 | Gstp1 | 59.54 | 67 | 7 |
| 2 | Sox2 | 50.12 | 50 | 36 |
| 3 | Krt8 | 45.62 | 67 | 32 |
| 4 | S100a6 | 44.20 | 64 | 38 |
| 5 | Elf3 | 39.94 | 69 | 28 |
| 6 | Gm26917 | 39.18 | 59 | 24 |
| 7 | Marcksl1 | 33.91 | 39 | 56 |
| 8 | Thbs1 | 32.50 | 66 | 25 |
| 9 | Tnfrsf12a | 31.61 | 73 | 22 |
| 10 | S100a1 | 31.36 | 87 | 18 |
| 11 | Prss23 | 30.75 | 77 | 17 |
| 12 | Fam162a | 28.60 | 43 | 46 |
| 13 | Nupr1 | 27.44 | 66 | 32 |
| 14 | Ereg | 27.29 | 64 | 15 |
| 15 | Nme2 | 23.88 | 77 | 6 |
| 16 | Scand1 | 22.29 | 48 | 53 |
| 17 | Lox | 21.54 | 54 | 39 |
| 18 | AY036118 | 21.30 | 30 | 57 |
| 19 | Txnip | 21.05 | 51 | 46 |
| 20 | Dmrtc2 | 20.41 | 27 | 63 |
| ★ | Stimulus 1 | 61.71 | 36 | 64 |
| ★ | Stimulus 2 | 40.75 | 45 | 23 |

**D** Top 20 regulated genes

| # | Target | Σ\|w\| | act. | inh. |
|---|---|---|---|---|
| 1 | Gm26917 | 28.14 | 58 | 34 |
| 2 | Cdkn2a | 25.07 | 37 | 54 |
| 3 | AY036118 | 23.99 | 59 | 34 |
| 4 | Gstp1 | 23.22 | 40 | 37 |
| 5 | Scand1 | 22.12 | 58 | 34 |
| 6 | Bgn | 21.08 | 42 | 47 |
| 7 | Nme2 | 20.99 | 29 | 42 |
| 8 | Gabarap | 20.73 | 47 | 47 |
| 9 | Ypel3 | 20.30 | 45 | 50 |
| 10 | Id3 | 20.05 | 42 | 54 |
| 11 | Marcksl1 | 19.40 | 63 | 36 |
| 12 | Egr1 | 19.21 | 48 | 44 |
| 13 | Xist | 19.19 | 56 | 38 |
| 14 | Tax1bp3 | 18.96 | 46 | 50 |
| 15 | Sdc4 | 18.91 | 63 | 34 |
| 16 | Dmrtc2 | 18.66 | 34 | 28 |
| 17 | Cdkn2b | 18.49 | 39 | 51 |
| 18 | 1110008F13Rik | 18.43 | 57 | 33 |
| 19 | Hist1h2ap | 18.41 | 57 | 36 |
| 20 | Mybl2 | 18.13 | 56 | 38 |

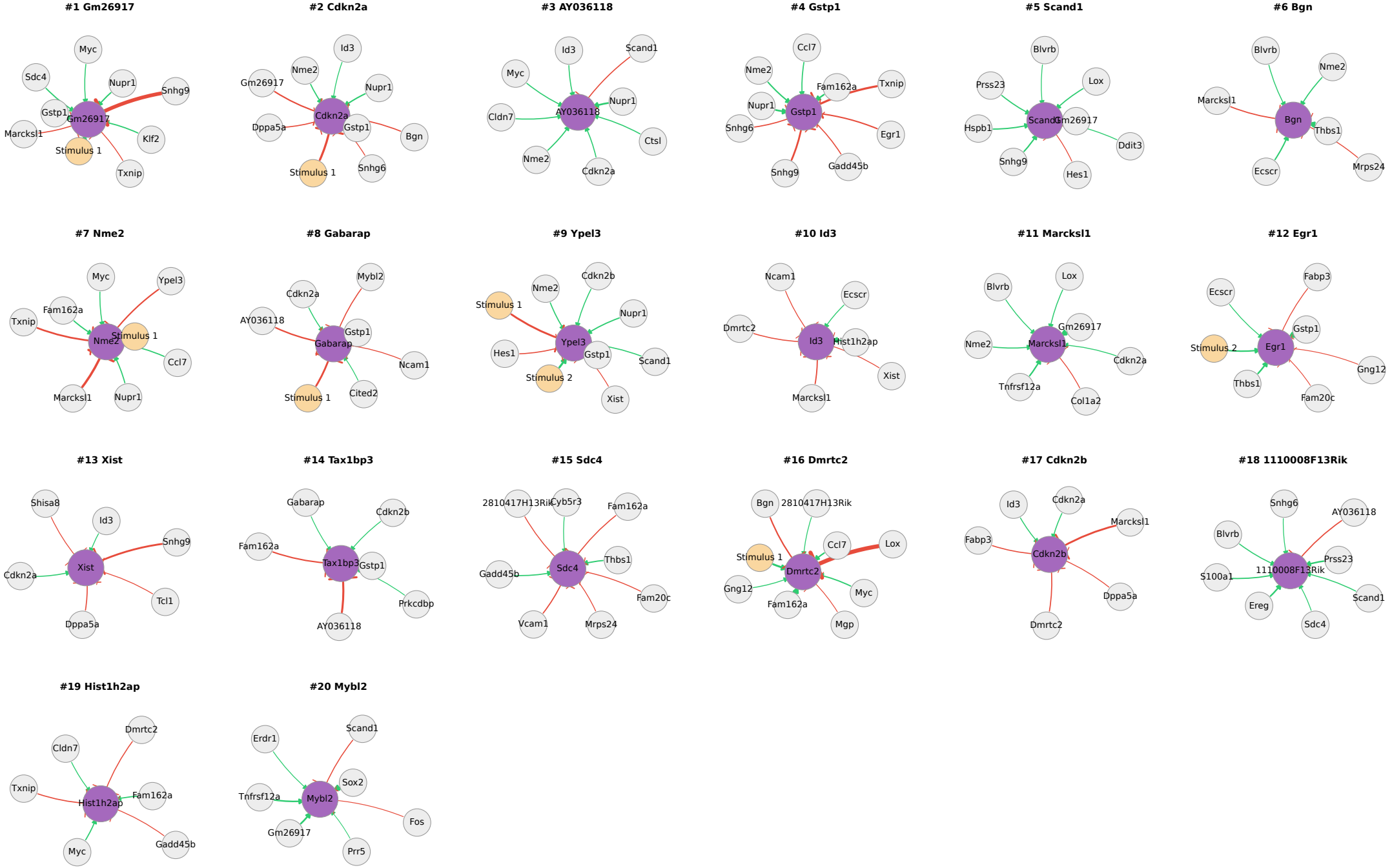
# 2. Gene regulatory network — top regulators and their main targets

Edges shown only if among the 10 strongest (|w|) targets of the regulator AND the 10 strongest regulators of the target; edge width ∝ |weight| (global scale).
Perturbed genes (KO/OV) are highlighted in beige.



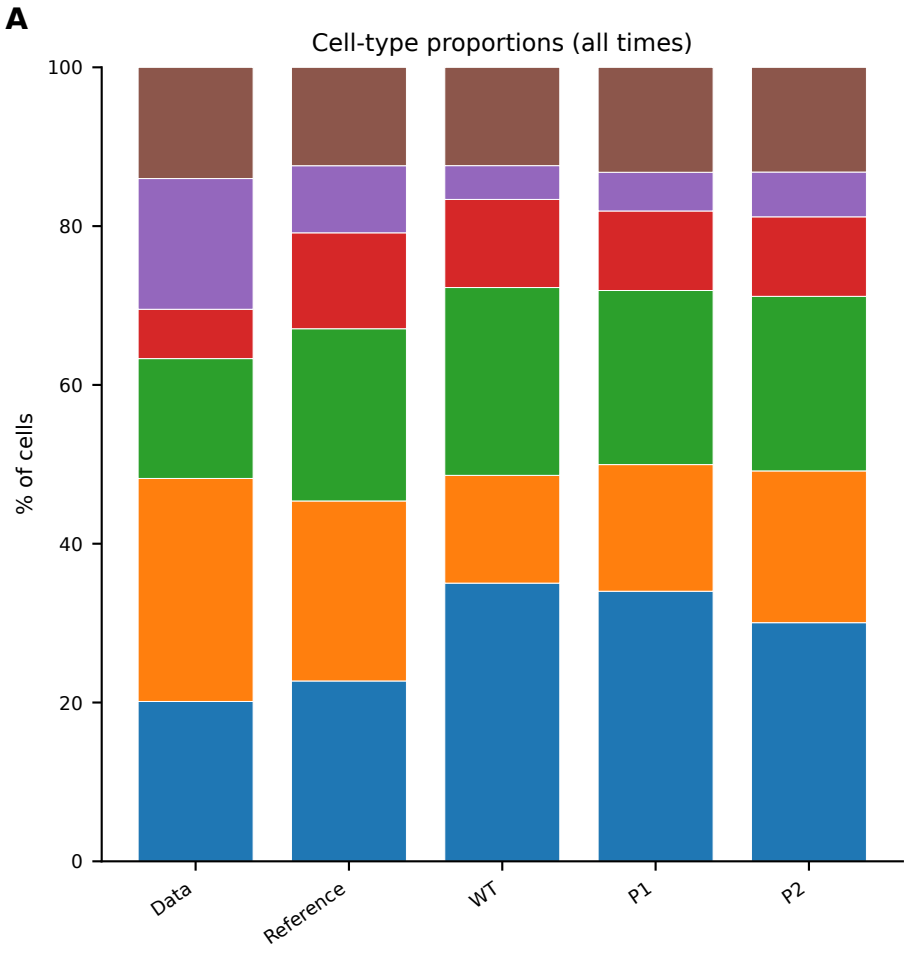
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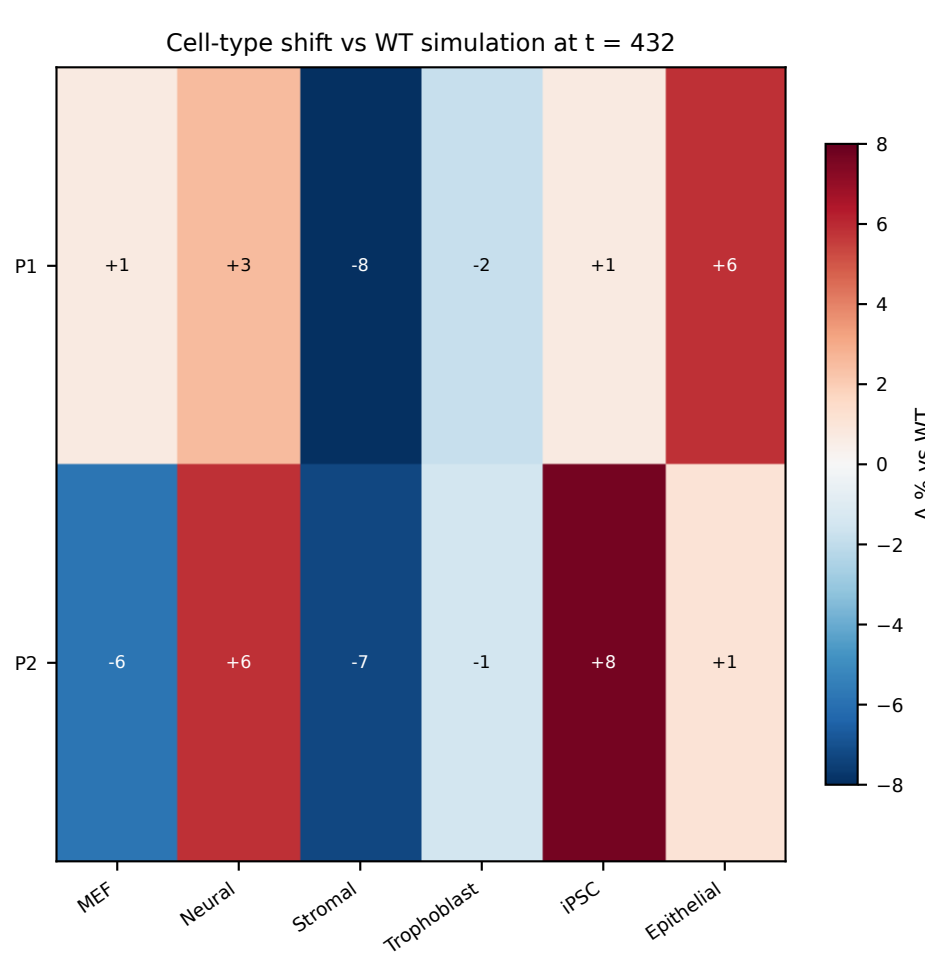
Edges shown only if among the 10 strongest (|w|) targets of the regulator AND the 10 strongest regulators of the target; edge width ∝ |weight| (global scale).
Perturbed genes (KO/OV) are highlighted in beige.



# 3. In-silico perturbations — overview

Cell types predicted by per-sample classifiers (logistic) trained on the observed data.



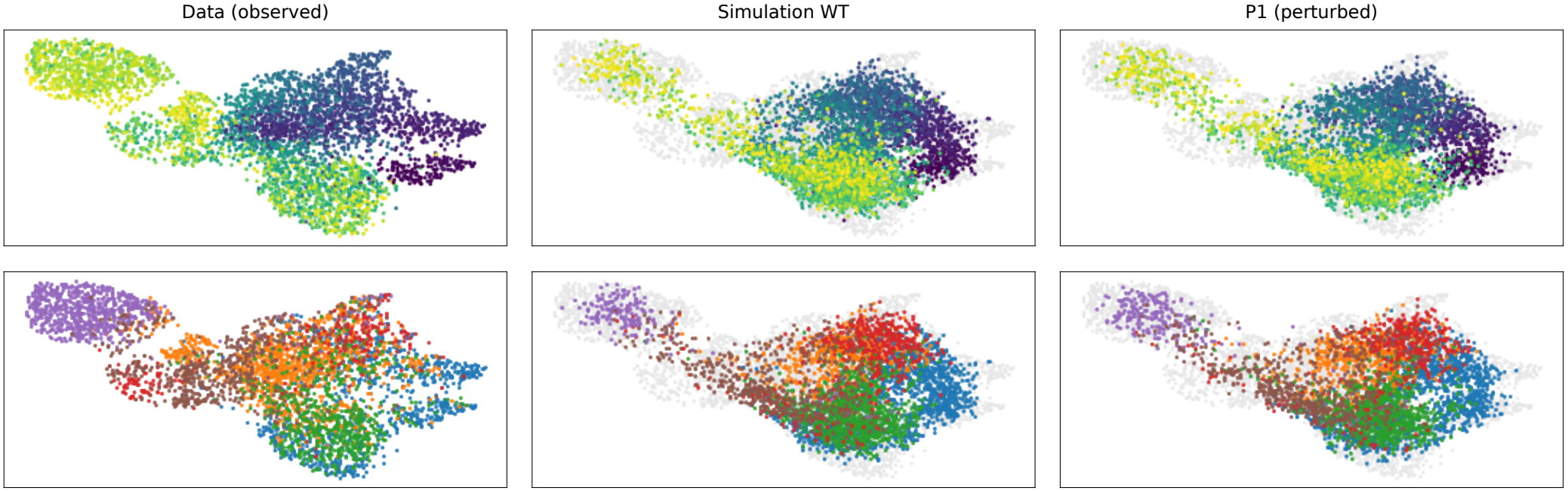

P1 OV Obox6

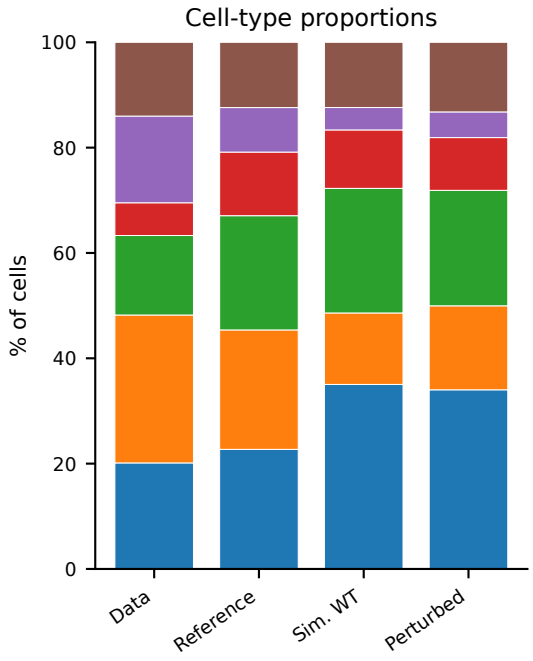
P2 OV Zfp42

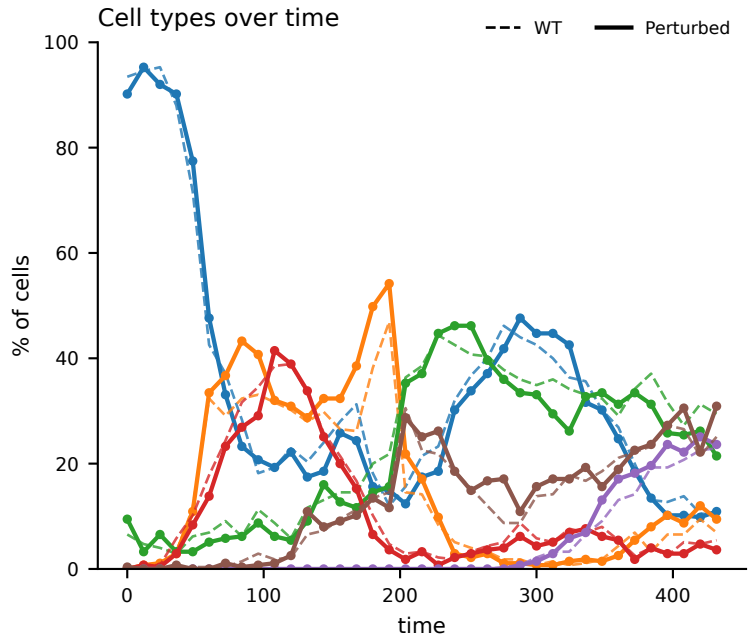
MEF Neural Stromal Trophoblast iPSC Epithelial

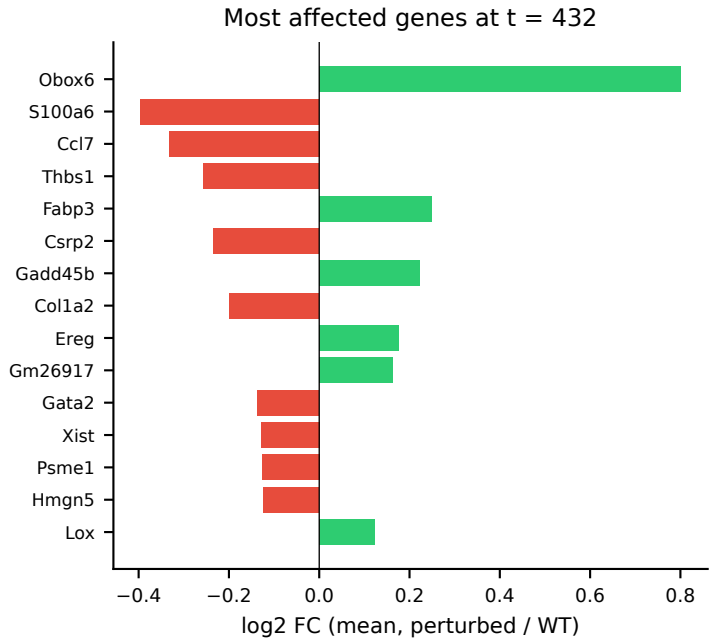
# 3. Perturbation P1 — OV Obox6

KO_none_OV_Obox6. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



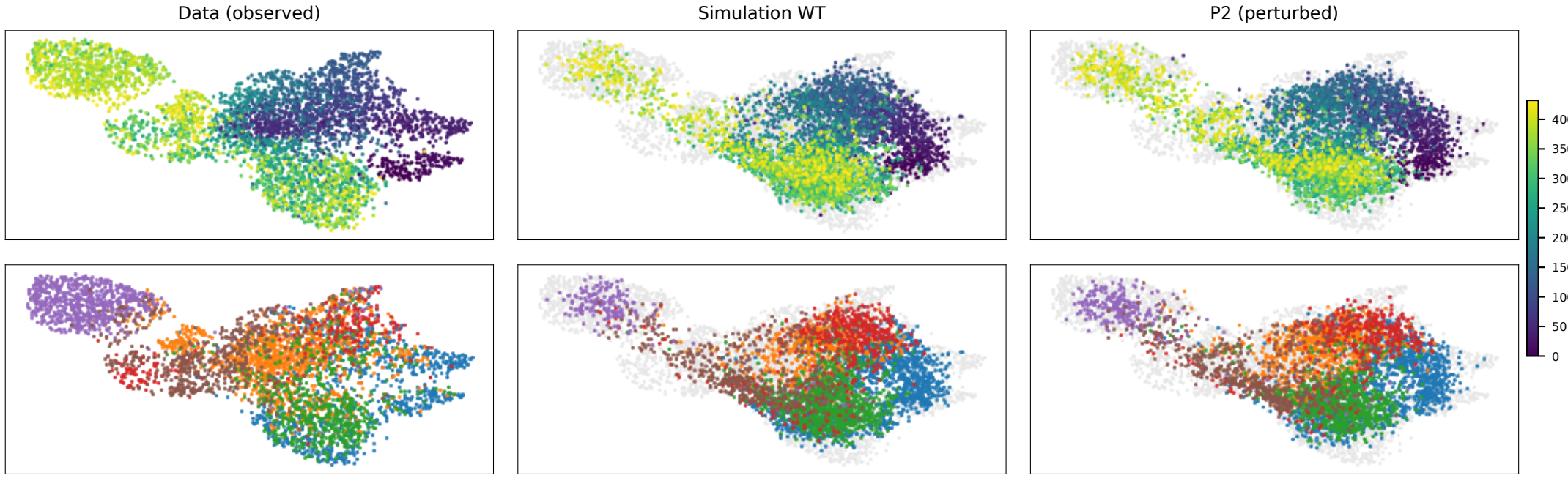



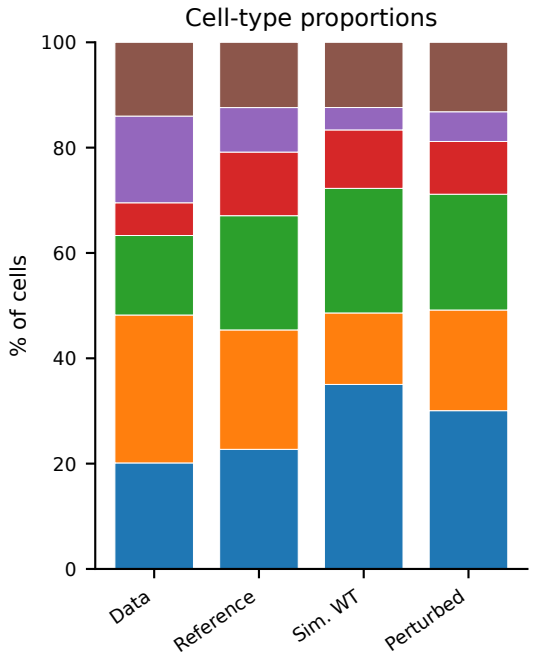



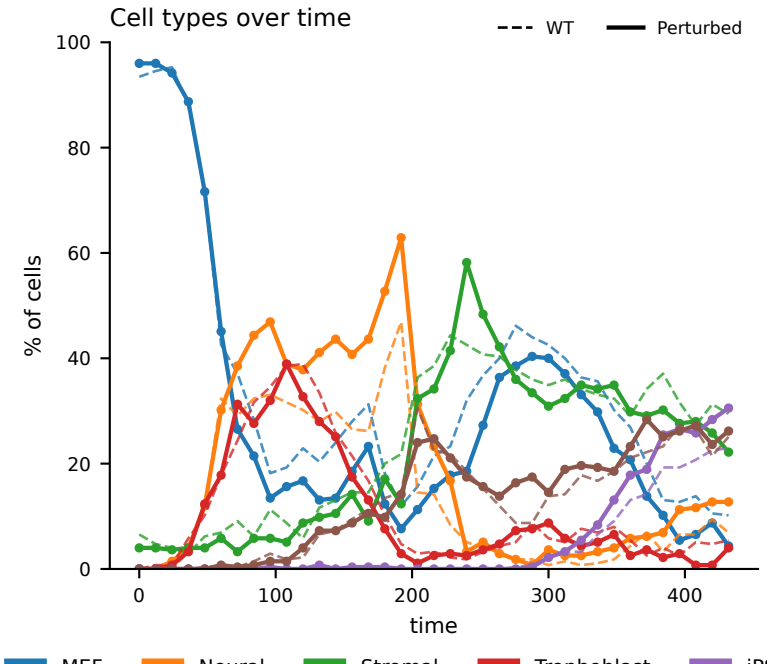

MEF Neural Stromal Trophoblast iPSC Epithelial

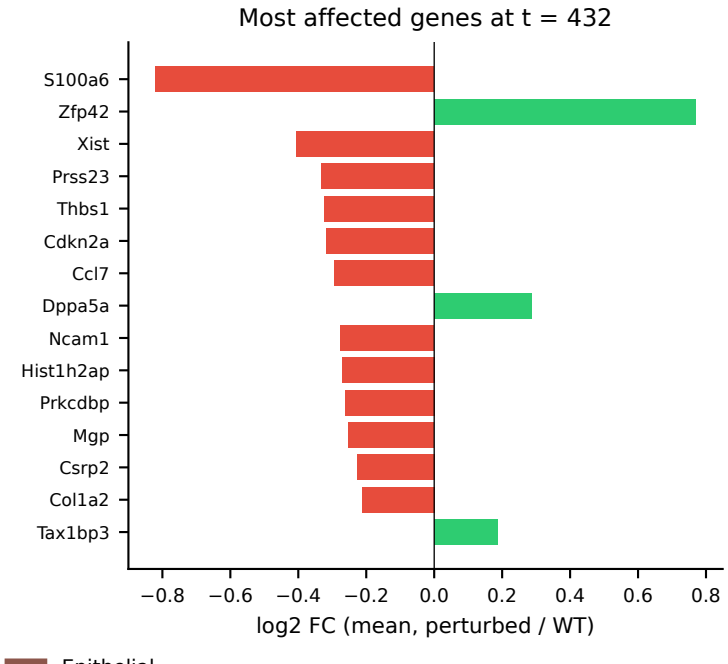
# 3. Perturbation P2 — OV Zfp42

KO_none_OV_Zfp42. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



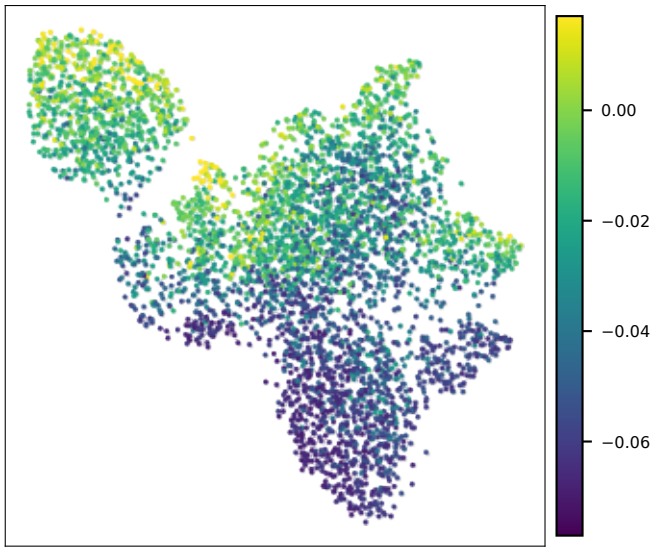





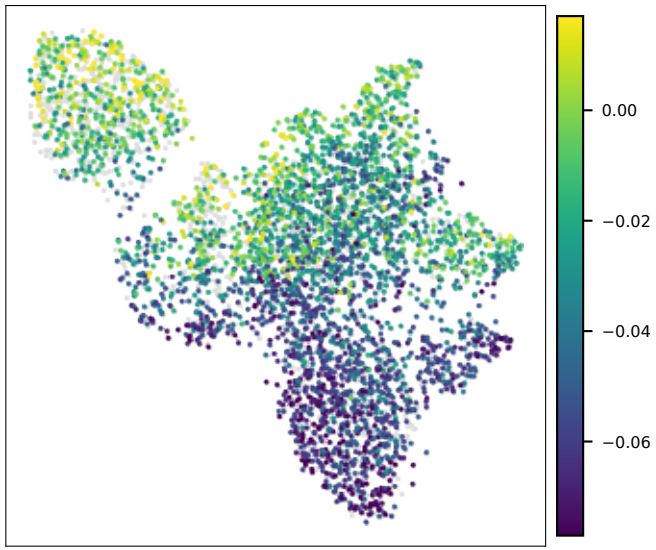

# 4. Proliferation — prior vs learned net rate (birth − death)

Learned = growth OT pass, mean over the trajectory states behind each observed cell (grey = cell not reached). Rates per time unit of the data.

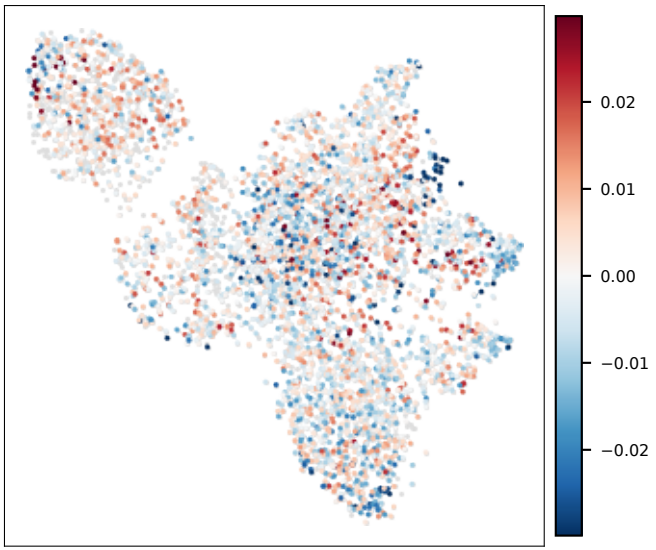
Prior net rate



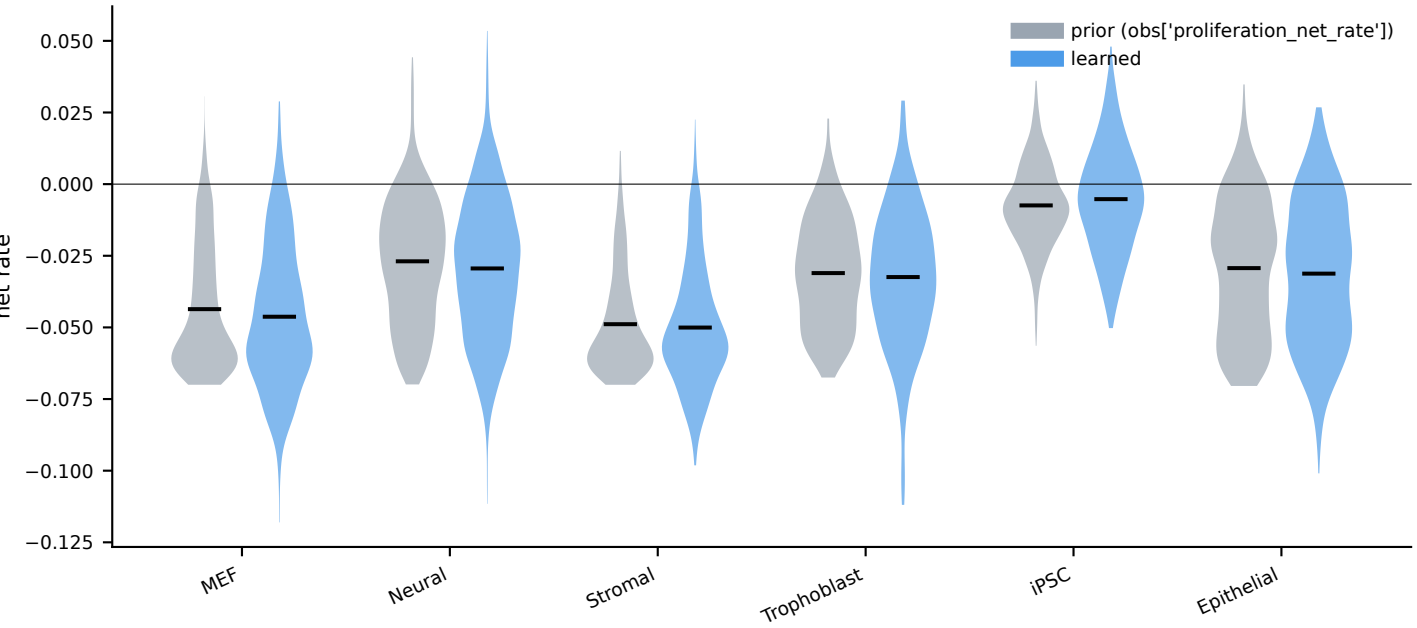

Learned net rate (growth OT pass)




Learned − prior




Net proliferation rate per cell type (cell_type)

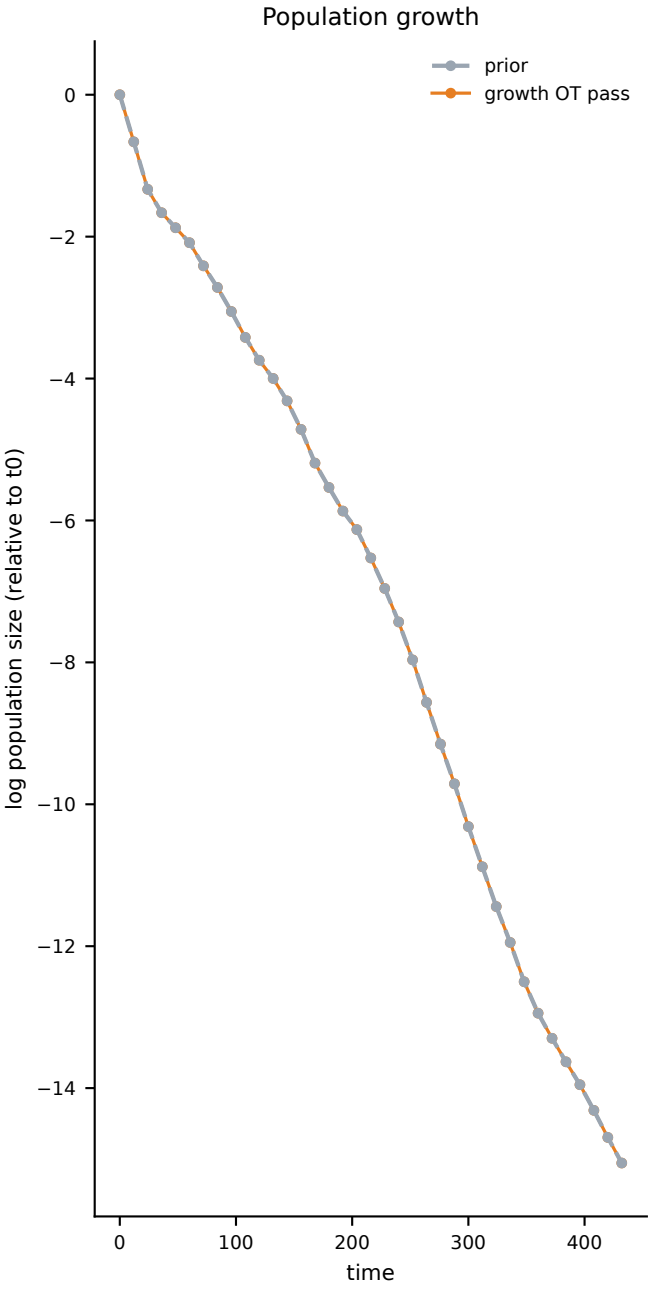


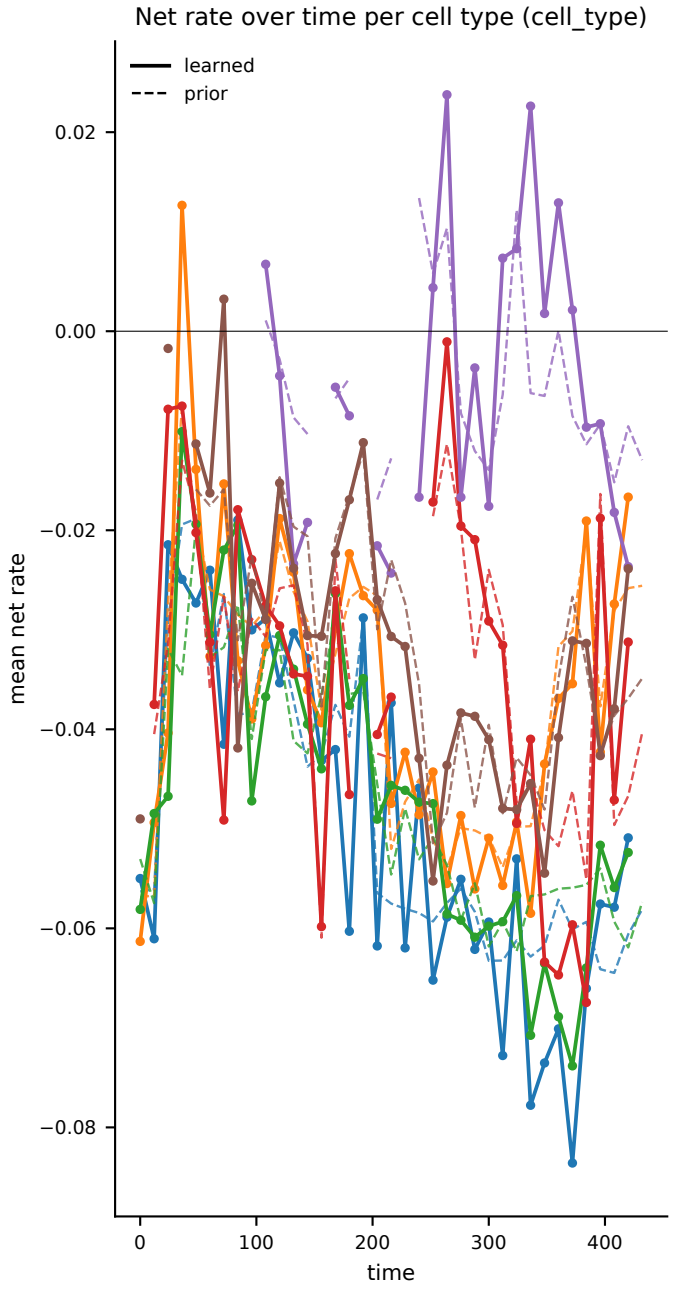

Mean net rate per cell type (cell_type)

| Cell type | Anchor | Prior | Learned |
|---|---|---|---|
| MEF | — | -0.0436 | -0.0463 |
| Neural | — | -0.0269 | -0.0294 |
| Stromal | — | -0.0489 | -0.0501 |
| Trophoblast | — | -0.031 | -0.0324 |
| iPSC | — | -0.00743 | -0.00524 |
| Epithelial | — | -0.0293 | -0.0312 |

# 4. Proliferation — population growth and drivers

Relative population size from the mean growth factor per interval; its absolute level follows the prior (anchored on the sheet population_sizes by fit_population_anchors, if filled).



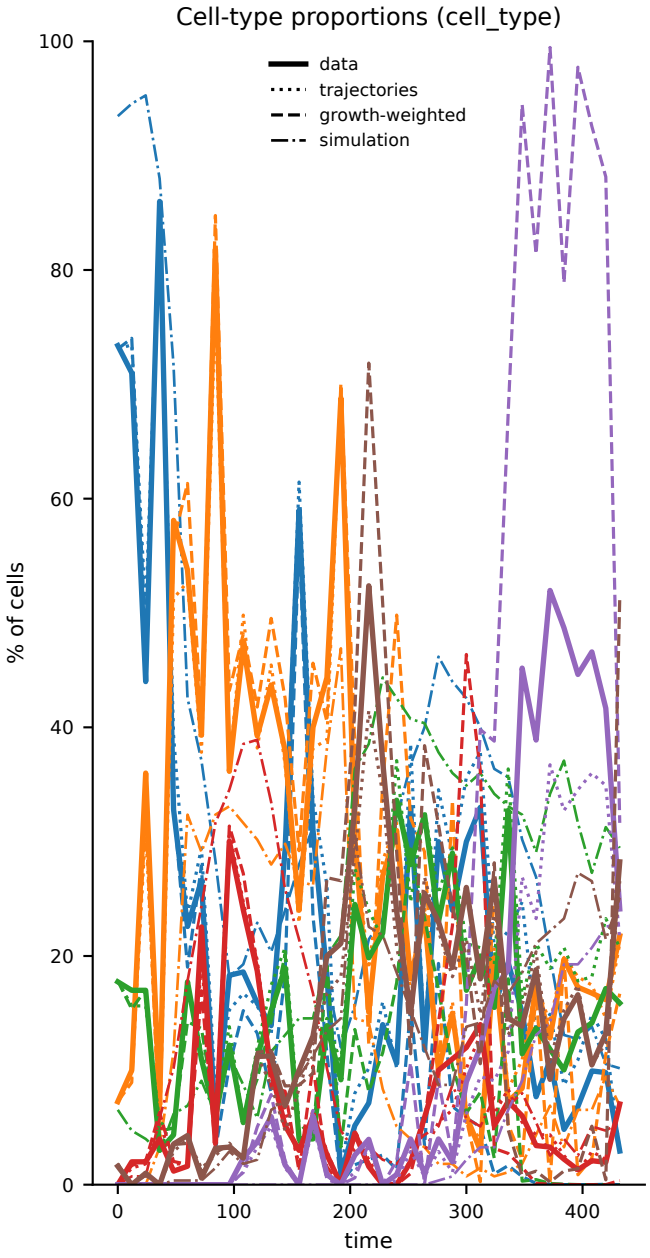



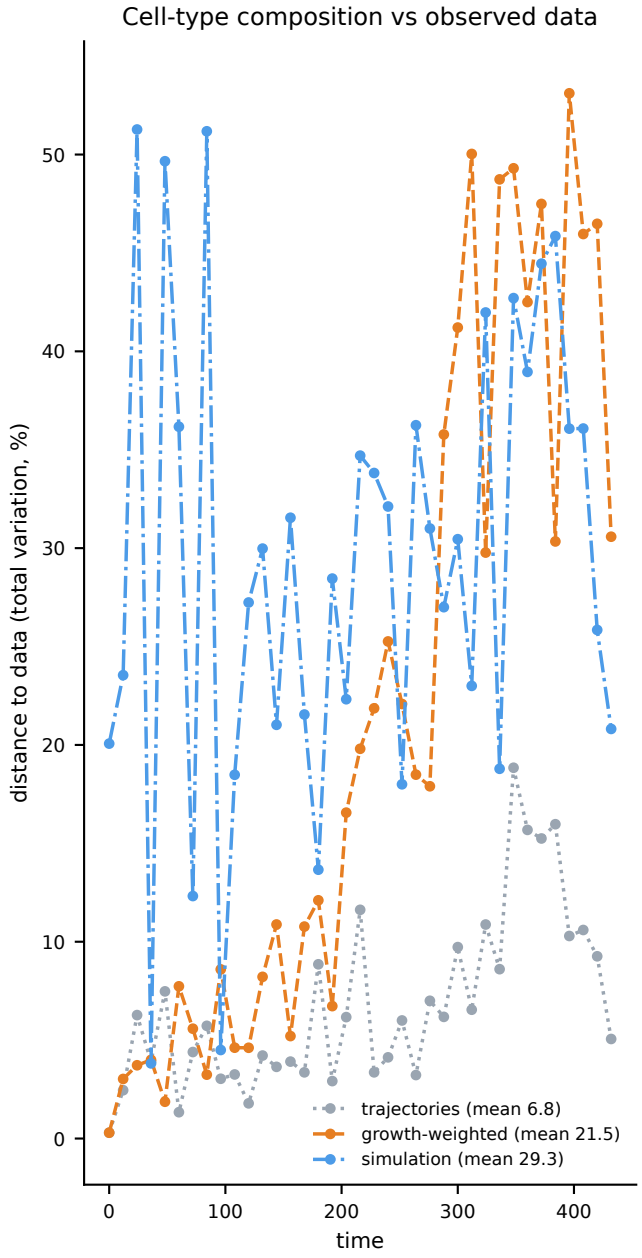

No proliferation MLP
(simulate_with_proliferation = True,
then infer_network_simul)

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# 4. Proliferation — growth-weighted trajectories and MLP fit

The OT selection keeps one descendant per ancestor (trajectories without proliferation); weighting each path by its expansion exp(∫R_opt) should bring them back to the observed data.




No MLP fit diagnostics
(rerun infer_network_simul
with proliferation)

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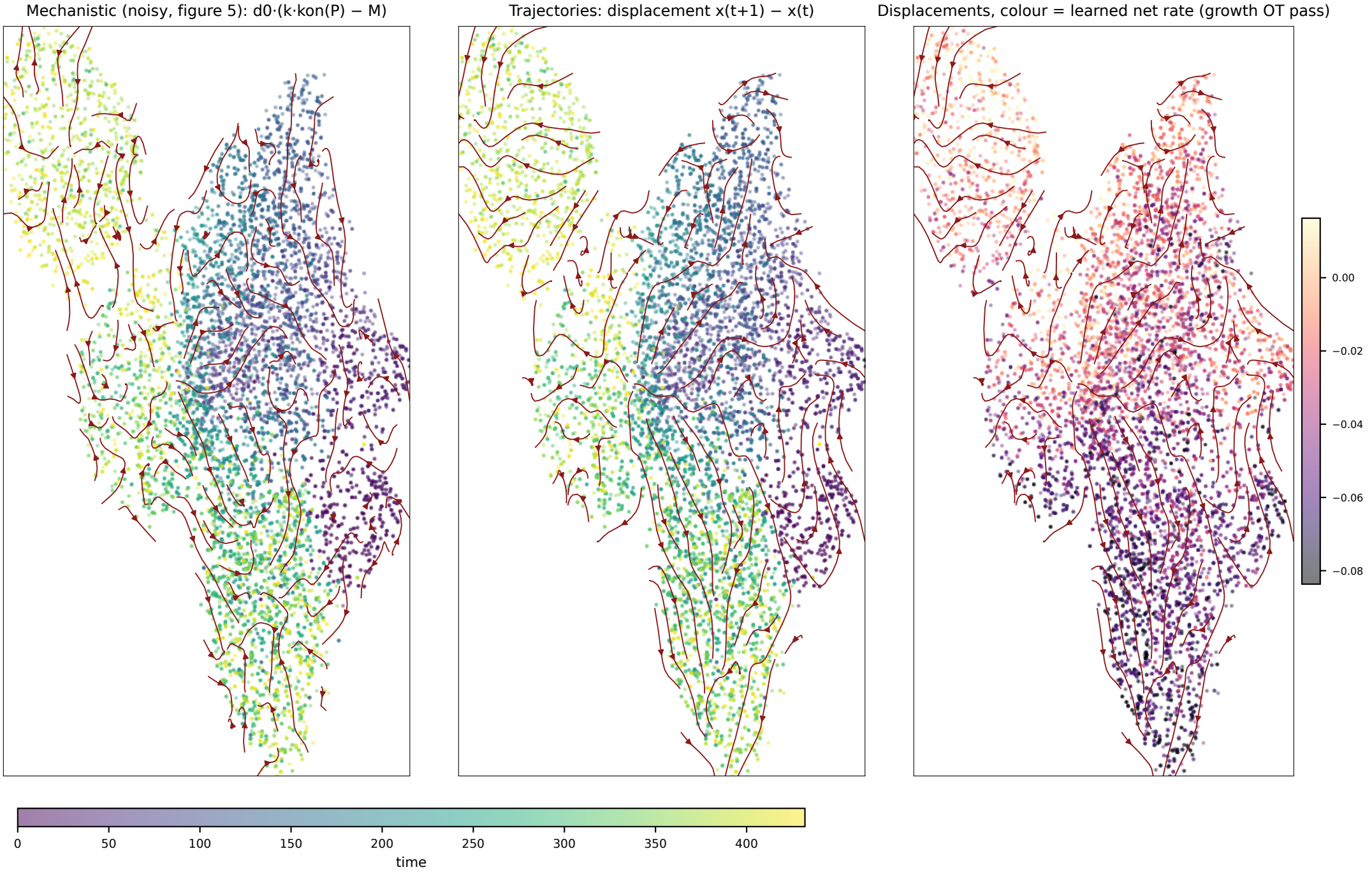
# 4. Protein dilution

**Section unavailable:**

```
ValueError('protein_dilution = False, or no birth rate (obs proliferation_birth_rate / net rate)')
```

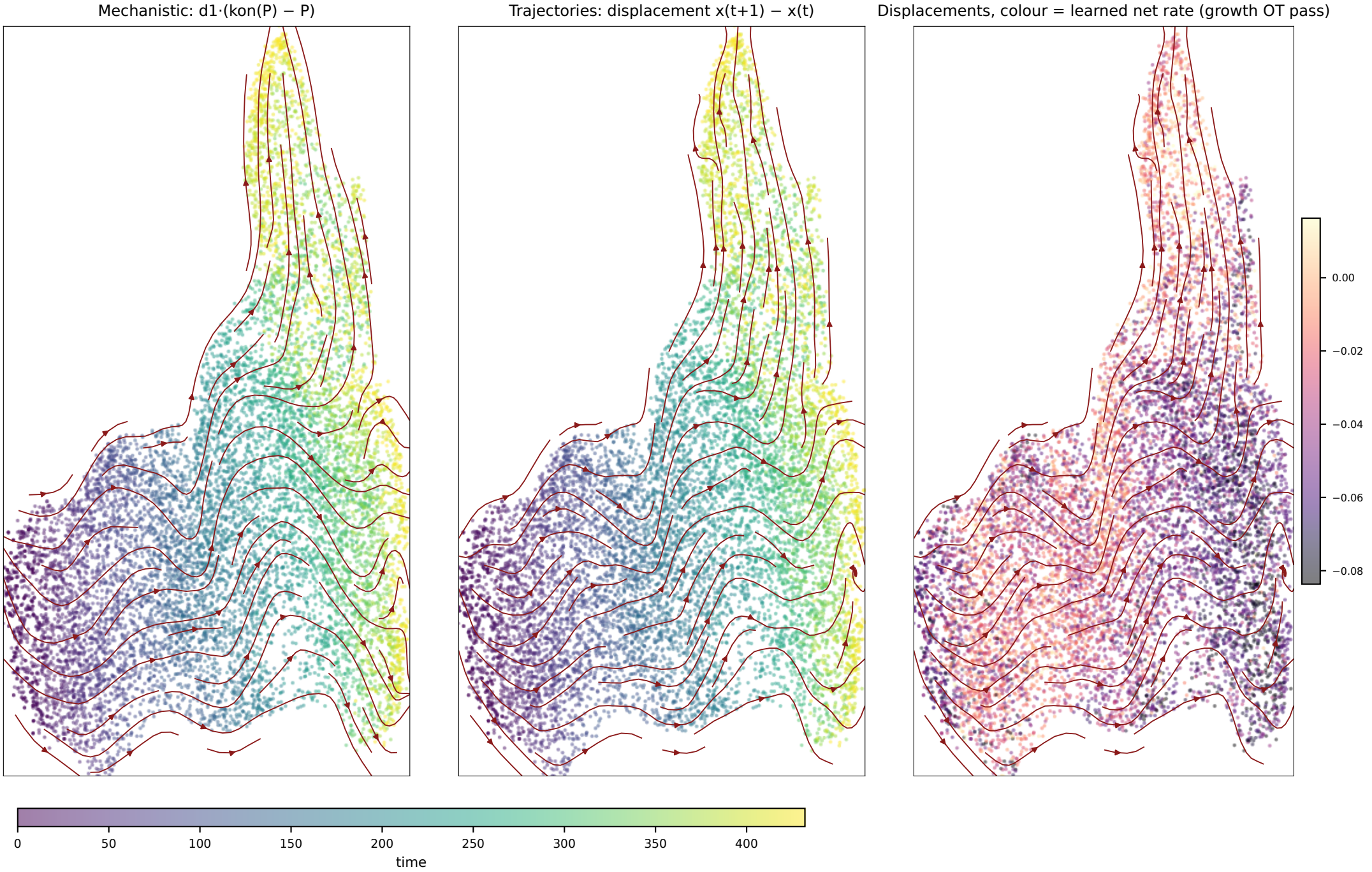
# 5. Learned dynamics — mRNA velocity and displacement fields

All 10175 trajectory states projected onto the UMAP learned on the observed mRNA (log1p, reference depth); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.08.



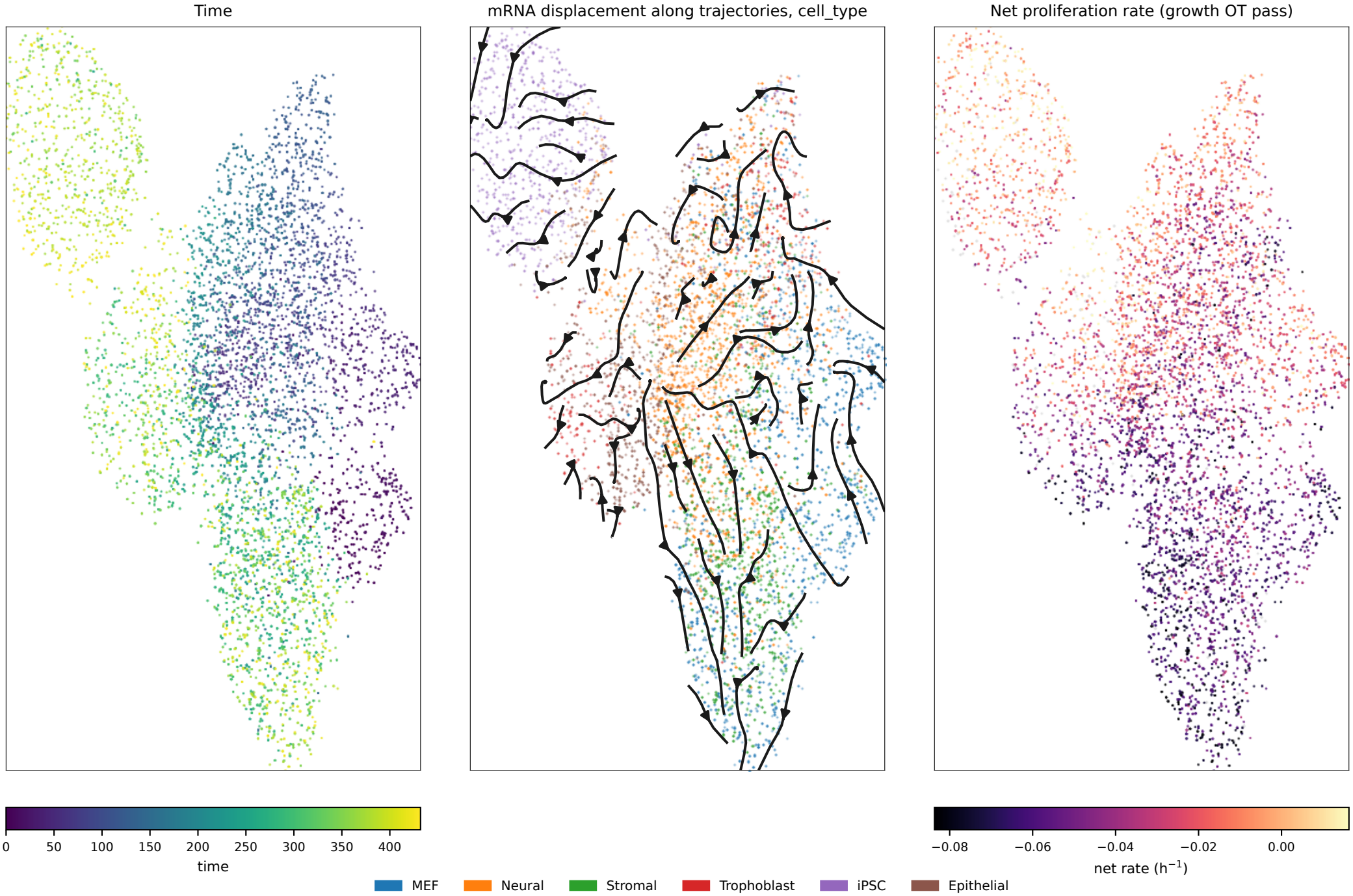
# 5. Learned dynamics — Proteins velocity and displacement fields

All 10175 trajectory states projected onto the UMAP learned on the protein trajectories (raw levels); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.49.



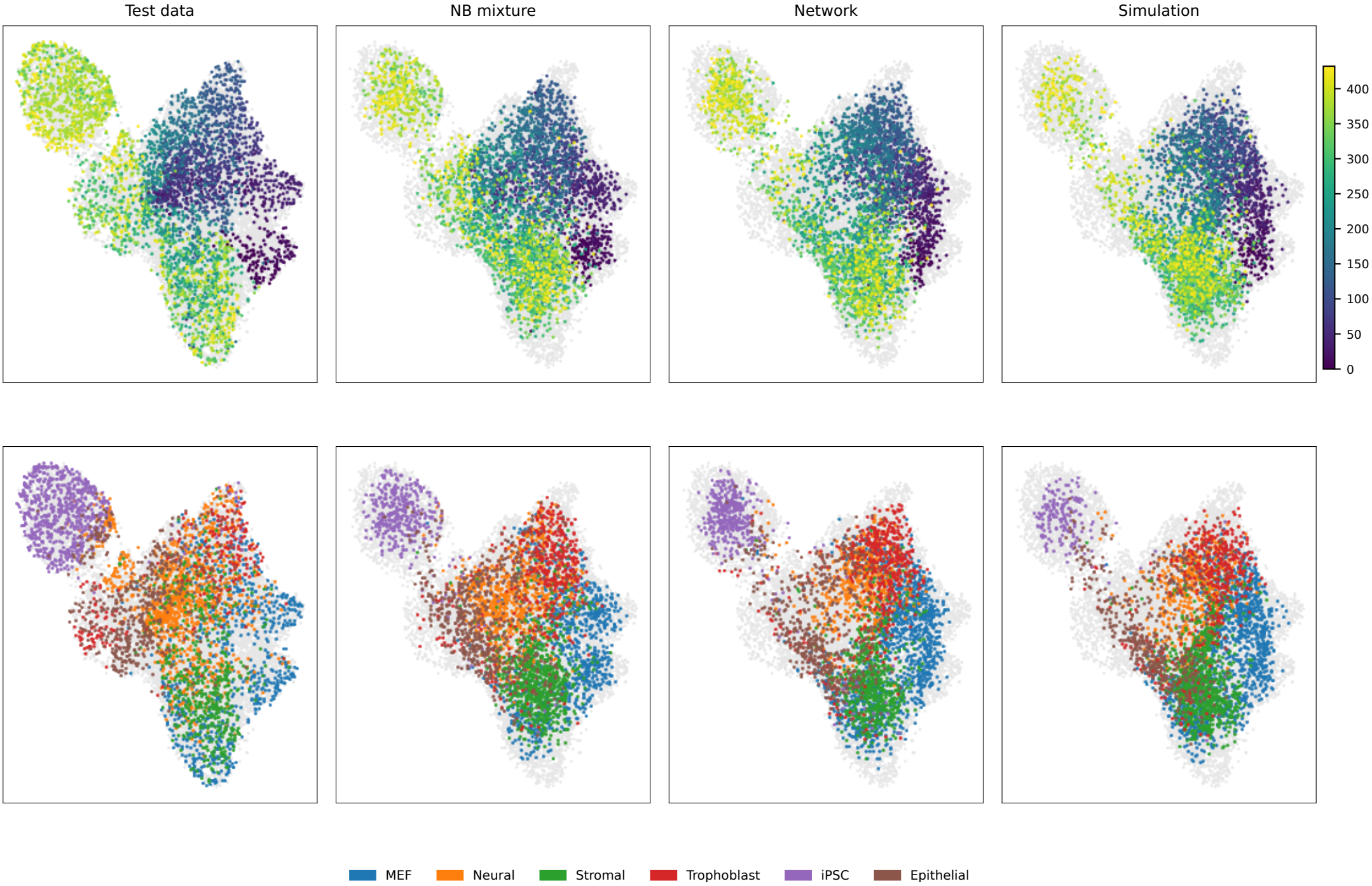
# 5. Learned dynamics — summary

All 10175 trajectory states on the UMAP learned on the observed mRNA (log1p, reference depth); displacement along the inferred trajectories x(t+1) − x(t); background of the field: cell_type.



# 6. Held-out test cells — predictions with the network fixed

Test cells classified into basins with the training mixtures, trajectories inferred with the training network fixed, then simulated; projected onto the UMAP learned on the observed data (grey: training).



# 6. Held-out test cells — quantitative checks

Per gene: 1-D Wasserstein distance between log1p counts, averaged over the common timepoints.

